

STATISTICAL ANALYSIS PLAN

A Phase 3 Double Blind Safety and Efficacy Study of Long-Acting Injectable Cabotegravir Compared to Daily Oral TDF/FTC for Pre-Exposure Prophylaxis in HIV- Uninfected Women

HPTN 084

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LIST OF ABBREVIATIONS

Term/Abbreviation	Definition
AE	Adverse Event
AIDS	Acquired immunodeficiency syndrome
ART	Antiretroviral therapy
BMI	Body Mass Index
CASI	Computer Assisted Self Interview
CRF	Case Report Form
CSR	Clinical Study Report
DAIDS	Division of AIDS
DMPA	Medroxyprogesterone
DSMB	Data and Safety Monitoring Board
EOS	End of Study
HPTN	HIV Prevention Trials Network
ITT	Intent to Treat
LA	Long Acting
NET-EN	Etonogestrel
NIAID	National Institute of Allergy and Infectious Diseases
OBSP	On Blinded Study Product
PYR	Person-Years
SDMC	Statistics and Data Management Center
SAE	Serious Adverse Event
SMC	Study Monitoring Committee
SOC	System Organ Class
TDF	tenofovir disoproxil fumarate
TDF/FTC	tenofovir/emtricitabine (trade name: Truvada®)
TFV	tenofovir
TFV-DP	tenofovir diphosphate

1. INTRODUCTION

This statistical analysis plan (SAP) details the statistical procedures that address the primary study objectives specified in Version 2.0 of Protocol HPTN 084 (dated: 6 November 2019): ***A Phase 3 Double Blind Safety and Efficacy Study of Long-Acting Injectable Cabotegravir Compared to Daily Oral TDF/FTC for Pre-Exposure Prophylaxis in HIV-Uninfected Women***. Any meaningful changes or additions to this SAP (e.g., in response to protocol amendments or violations of assumptions underlying pre-planned analyses), and the timing of such changes in relation to unblinding of statisticians performing the analyses, will be documented in a designated section of the SAP. New versions of the SAP will be issued to document updates and changes in the plan. Analysis plans for sub-studies not addressed in the protocol, and for secondary analyses not anticipated prior to study completion, and for exploratory analyses will be developed as separate documents.

Plans for formal interim analysis of trial data are outlined in Section 7.2. A more detailed plan is documented separately in HPTN 084 Interim Monitoring Guidelines. Any substantive modifications to analysis methods or type I error control made in response to DSMB reviews or changes in the protocol will be documented in subsequent versions of the SAP. In DSMB reviews, the HPTN Statistical and Data Management Center (SDMC) will routinely report on operational metrics (e.g., rates of recruitment, retention, study drug discontinuation). These reports are also routinely shared with the study sites to monitor operational performance based on these metrics.

2. GENERAL DESIGN CONSIDERATIONS

2.1 Purpose:

To evaluate the safety and efficacy of the injectable agent, cabotegravir (CAB LA) compared to daily oral tenofovir disoproxil fumarate/emtricitabine (TDF/FTC), for pre-exposure prophylaxis (PrEP) in HIV-uninfected women.

2.2 Design:

Multi-site, double blind, two-arm, randomized (1:1), controlled superiority trial of the safety and efficacy of CAB LA compared to daily oral TDF/FTC for HIV prevention.

2.3 Population:

HIV-uninfected women at risk for acquiring HIV, 18 to 45 years old.

2.4 Study Size:

The original sample size calculation was to enroll 3200 women. In response to a recommendation from the HPTN Study Monitoring Committee in October 2020, this target was increased to 3350 in LOA 3.

2.5 Study Duration:

Approximately 4.6 years total, with individual participants being followed on randomized study product between 1.6 years (for the last enrolling participants) to approximately 3.6 years (for the earliest enrolling participants), and on oral TDF/FTC for an additional 48 weeks. Accrual will require approximately 2 years.

2.6 Study Sites:

Study sites will be in sub-Saharan Africa (SSA).

2.7 Study Regimen and Follow-up:

Once randomized to one of two arms, participants will move through the following steps (active study products are shown in bold text):

2.7.1. Step 1, Oral Run-in Phase:

Arm A – **Daily oral cabotegravir (CAB)** and oral TDF/FTC placebo for five weeks plus an HIV prevention package including behavioral risk reduction and adherence counseling, provision of condoms.

Arm B – **Daily oral TDF/FTC** and oral CAB placebo for five weeks plus an HIV prevention package including behavioral risk reduction and adherence counseling, provision of condoms.

Individuals who discontinue study participation after receiving oral product in Step 1, but do not receive any study product in Step 2 will be offered the option to be followed annually with HIV testing until study end.

A participant that becomes HIV-infected during Step 1 of the study will permanently discontinue study product, be terminated from the study, and referred for HIV-related care.

2.7.2 Step 2, Injection Phase:

Arm A – **CAB LA** as a single intramuscular [IM] injection at two time-points four weeks apart and every eight weeks thereafter and daily oral TDF/FTC placebo plus an HIV prevention package including behavioral risk reduction and adherence counseling, provision of condoms.

Arm B – **Daily oral TDF/FTC** and IM placebo (matching vehicle, identical volume as active injectable product in Arm A) at two time-points four weeks apart and every eight weeks thereafter plus an HIV prevention package including behavioral risk reduction and adherence counseling, provision of condoms.

Step 2 will continue until the required number of endpoints (114) is reached, estimated to be 81 weeks after enrolling the last participant.

Participants who prematurely discontinue blinded study products during Step 2 (having received at least one injection) for any reason other than HIV infection will be offered to transition to 48 weeks of open-label TDF/FTC followed by annual testing through the duration of step 3.

A participant who becomes HIV-infected during Step 2 of the study will permanently discontinue study product, be placed on immediate suppressive ART, and will be facilitated into locally provided ART, and followed at quarterly intervals for 48 weeks after

their last injection prior to diagnosis of HIV in order to test for safety parameters, as well as CD4 cell count and HIV viral load. After 52 weeks, they will be terminated from the study and transitioned to continued HIV-related care.

2.7.3 Step 3, Follow-up Phase:

Arms A and B – Open-label daily TDF/FTC (in order to cover the pharmacokinetic [PK] tail for Arm A participants) will be provided no later than eight weeks after the last injection visit, for up to 48 weeks plus an HIV prevention package including behavioral risk reduction and adherence counseling, and provision of condoms. Participants will then transition to locally available HIV prevention services, including services for PrEP, if available.

Participants with confirmed HIV infection during Step 3 will have their TDF/FTC stopped and be followed quarterly at least for the duration of Step 3, with possible additional assessments and follow-up to be determined by guidance from members of the HIV alias. Study product will be discontinued and participants will be referred for care.

2.7.4. Pregnancy

A participant with a first positive pregnancy test will temporarily discontinue blinded injections and receive a 4-week supply of open label TDF/FTC and may resume blinded treatment if a second test after 4 weeks indicates that she is not pregnant. All pregnant participants with a confirmed positive pregnancy test (four weeks after the initial pregnancy test) will be unblinded and followed by the study every 12 weeks. Regardless of the randomization assignment or point in the study, all pregnant participants will be placed on open-label TDF/FTC for the duration of the pregnancy. Once pregnancy outcome is reached, and after the participant is not breastfeeding, she may resume study product, according to original randomized arm but open-label, and resume visits per the study schedule.

If participants decline to use an approved contraceptive method, they will be given open-label TDF/FTC for 48 weeks starting no later than eight weeks after her last CAB LA/placebo injection. If study follow-up is ongoing she will be required to be followed up at least annually for HIV testing thereafter. If a participant transitioned to open-label TDF/FTC during Step 2, either because she wanted to become pregnant or declined to continue using long-acting contraception, and subsequently changes her mind she may continue in the study on blinded study product (per original randomization) once she has documentation of a negative pregnancy test and of resuming long-acting contraception.

3. STUDY OBJECTIVES

3.1. Primary Objective:

- **Efficacy:** To evaluate the relative efficacy of oral CAB/CAB LA (oral run-in and injections, Steps 1 and 2) vs. daily oral TDF/FTC for HIV prevention (Steps 1 and 2).
- **Safety:** To evaluate the relative safety of oral CAB/CAB LA (oral run-in and injections, Steps 1 and 2) vs. daily oral TDF/FTC for HIV prevention (Steps 1 and 2).

3.2. Secondary Objectives:

- To compare HIV incidence among participants receiving oral CAB/CAB LA vs. daily oral TDF/FTC (Steps 1, 2 and 3).
- To evaluate relative efficacy of oral CAB/CAB LA vs. oral TDF/FTC in subgroups defined by the baseline factors of age, herpes simplex virus-2 (HSV-2) serostatus, contraceptive method, and body mass index (BMI).
- To describe and model the relationship between HIV incidence and drug concentration, within each arm.
- To describe the distribution and correlates of drug concentration, within each arm.
- To compare the acceptability of and preferences for CAB LA vs. oral TDF/FTC.

3.3. Tertiary Objectives:

- To estimate sexual risk behaviors, as measured by self-report and rates of incident sexually transmitted infections (STIs).
- To compare Grade >2 adverse event (AE) rates in women with baseline BMI $\leq$ 30 kg/m², within each study arm.
- To compare differences in weight gain and BMI, by arm.
- To compare pregnancy incidence and outcomes between arms.
- To evaluate rates of HIV drug resistance among participants who acquire HIV infection during the study among participants receiving oral CAB/CAB LA vs. oral TDF/FTC.
- To determine plasma concentrations of medroxyprogesterone (DMPA) or norethisterone (NET-EN) or etonogestrel when co-administered with CAB LA.

4. STATISTICAL CONSIDERATIONS

4.1 Sample Size and Power

The primary analysis will be based on HIV incidence during Steps 1 and 2. As described previously, we assume that participants will be followed between 81 (latest enrollees) and 185 (earliest enrollees) weeks in Steps 1 and 2 (1.6 – 3.6 years), with a uniform distribution of enrollments over a two-year period. Thus, average time in Steps 1 and 2 will be 133 weeks (2.6 years). Sample size calculations are based on the following assumptions:

- Background HIV incidence, in the absence of any PrEP, is 3.5% per year
- Both CAB LA and TDF/FTC are 85% effective when used with 100% adherence
- 2.5% one-sided type I error rate and 90% power at the indicated alternative
- Average follow-up duration of 2.6 years (range: 1.6 – 3.6 years)
- Maximum 5% lost-to-follow-up per year

Table 4.1 presents five scenarios and associated total sample sizes. All are superiority designs. The first scenario assumes that adherence to TDF/FTC and CAB/CAB LA will be 50% and 85%, respectively, averaged over the entire Step 1 and 2 follow-up period. The second and third scenarios assume a higher adherence to TDF/FTC (second line) and lower adherence to CAB LA (80%) (third line) (these scenarios are considered unlikely). The last two scenarios retain the conservative assumption of 80% adherence to CAB LA and assume lower adherence to TDF/FTC of 45% and 48%. The largest blinded trials of TDF/FTC among women in a similar setting have shown even lower adherence than assumed in Table 4.1.

Given this history we believe that a sample size of at least 3,128 (111 events for a fixed sample size trial; we will target 114 events to allow for interim monitoring) provides an adequate degree of robustness against uncertainties in adherence rates to the two drug regimens. The trial will continue until we reach 114 events, or a stopping boundary is crossed (see below).

Table 4.1. Five Case Scenarios

Adherence		HIV Incidence (%/year) ¹				
TDF/FTC	CAB LA	TDF/FTC	CAB LA	RR	Number Events	Sample Size
.50	.85	2.01	0.97	.48	78	2352
.55	.85	1.86	0.97	.52	98	3112
.50	.80	2.01	1.12	.56	125	3590
.48	.80	2.07	1.12	.54	111	3128
.45	.80	2.16	1.12	.52	98	2686

¹Background incidence is reduced by a weighted average of fully adherent and non-adherent individuals e.g. $2.01 = 3.5 * ((1-.85) * .5 + 1 * (1-.5))$

This study will be monitored by a NIAID DSMB, which will meet at least annually to review safety and efficacy data, as well as data quality. The DSMB will use an **O'Brien-Fleming boundary** to consider stopping the trial early for efficacy.

4.1.1 Sample Size Calculation and Monitoring for Efficacy and Harm

Table 1 reviews the key sample size assumptions. We propose five analyses of the 084 data (four interim and one final). Figure 1 shows the times of the interim analyses and the expected accumulation of participants and information (HIV events) over the course of the trial (assuming a constant incidence). The times of the interim analyses are chosen to correspond to the times of the DSMB reviews (every six months). No interim analyses are proposed prior to 1.5 years because of the small amount of information (HIV events) expected early in the trial. The software nTerim 3.0 was used for calculations.

Table 1. Key sample size calculation parameters and results. EOS = end of steps 1 and 2.

Parameter	Value	Comment
Test Significance level, α (2 sided)	0.05	
Average duration of follow-up	133 wks	$(81 + 185)/2$
Proportion uninfected at EOS, Cab	.9718	$\text{Exp}(-.0112 * 133/52)$ (1.12%/yr incidence)
Proportion uninfected at EOS, TDF/FTC	.9484	$\text{Exp}(-.0207 * 133/52)$ (2.07%/yr incidence)
Hazard ratio	.54	Assuming constant hazard in each arm
Power	90%	

Number of events	114	Slightly inflated from fixed sample size 111 due to interim monitoring
Total sample size	3257	2834/.87 where overall retention = .87
Interim monitoring times (expressed as % information)	.22, .39, .59, .78, 1	Corresponds to 6-month intervals, starting at 1.5 years

Figure 1. Planned accumulation of sample size and statistical information, along with proposed times of interim analyses, based on HPTN 084 study design.

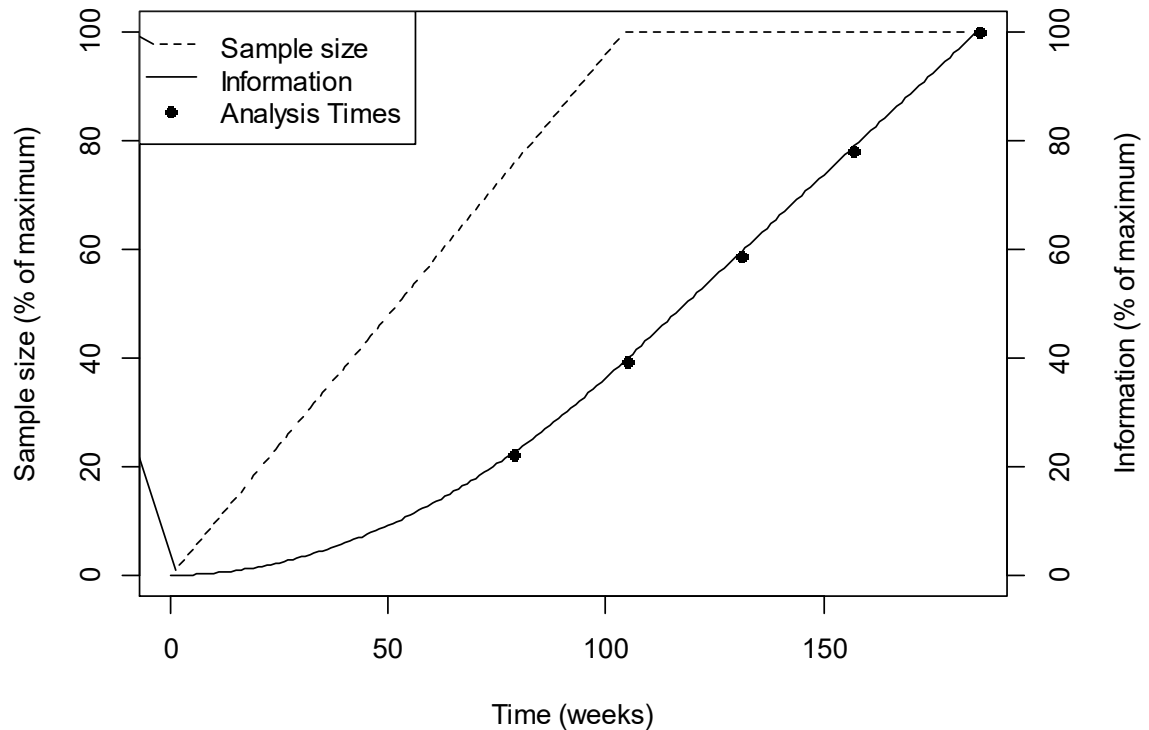


Table 2 provides detailed information on monitoring the trial for early stopping at the interim analyses. The O'Brien-Fleming spending function is used to determine stopping boundaries. By the time of the first or second analysis we should have a better understanding of enrollment rates and HIV incidence; the timing of the final 3-4 analyses may be modified at that time. Any modifications, if needed, will use the O'Brien-Fleming spending function applied to the cumulative information available at the time of the analysis.

Table 2. Interim analysis plan for monitoring HPTN 084 based on expected accrual and HIV incidence rates. Ho: RR = 1; Ha RR = .54. Type I error rates (α) in the table are for 2-sided tests.

Time (years)	% Information	Expected no. events	Efficacy boundary (Z)	Harm boundary (Z)	Efficacy boundary (RR) [#]	Nominal α	Cumulative α	Cumulative Stop Prob. under Ha
1.5*	22	25.3	-4.64	4.64	.04	.000001	.000001	0.1
2.0	39	44.8	-3.40	3.40	.30	.00066	.00066	8.7
2.5	59	67.0	-2.70	2.70	.50	.00682	.00704	42.7
3.0	78	89.2	-2.32	2.32	.60	.02010	.0223	72.2
3.56	100	114	-2.02	2.02	.68	.0429	.05	90.0

** Removing the first interim analysis has virtually no effect on the boundaries at subsequent times
RR boundary = (#events Cab/# events TDF), conditional on expected number of events; these RR correspond to approximately 1, 10, 22, 33, 46 events in the CAB arm, respectively*

4.2 Randomization

Participants will be randomized to one of two study arms in a 1:1 ratio. Randomization will be stratified by study site, and a permuted blocks design will be used to ensure balanced treatment assignments within study site. The randomization scheme will be generated, operationalized and maintained by the HPTN Statistics and Data Management Center.

4.3 Blinding

Study site staff, except for the site Pharmacist of Record or their designee, and participants will be blinded to the random assignments. Blinding will be maintained until the trial is completed or stopped, i.e., the trial is stopped early, or the required number of endpoints (114) has been met. At a specified time directed by the HPTN Statistics and Data Management Center, participants will be notified of their treatment assignment.

In addition, as described in sections 5.14 and 5.18 of the HPTN Protocol 084, v2.0, participants with a confirmed pregnancy will be unblinded. Participants who are unblinded due to pregnancy may restart open-label study product in their original randomization arm (open label CAB LA or TDF/FTC) following delivery and cessation of breastfeeding.

4.4 Accrual and Retention

Approximately 3,350 participants will be enrolled in approximately 24 months and followed through Steps 1 and 2 for 1.6 to 3.6 years and on oral TDF/ FTC for an additional 48 weeks. An average annual retention rate of at least 95% percent will be targeted (87 - 88% for the entire Step 1 and 2 follow-up period).

5. PROTOCOL DEVIATIONS

All protocol deviations will be reported and reviewed on a regular basis, and each case will be managed clinically and with regards to ongoing product administration by the CMC and DAIDS MOs during the course of the trial. Certain serious protocol deviations, identified during the blinded review, may warrant the exclusion of participants from the ITT or Per-Protocol analysis (guidance for adjudication for these deviations is provided in the HPTN 084 Deviation Adjudication Terms of Reference document). Generally, exclusions from the mITT analysis are expected to be limited to participants deemed inappropriately enrolled and resulting in termination from the study. Any exclusion must be carefully adjudicated and scientifically justified prior to breaking the blind for individuals involved in the adjudication process.

6. ANALYSIS POPULATIONS

6.1 Randomized Population

All participants who were randomized

6.2 Intent to Treat (ITT) Population

All participants who were randomized, excluding those who were subsequently found to be inappropriately enrolled due to violations of the inclusion/exclusion criteria that were deemed to warrant exclusion by the adjudication committee.

6.3 Modified ITT (mITT) Population

All women who are included in the ITT Population and who were not HIV infected at enrollment are included in the mITT population.

6.4 Per-Protocol (PP) Population

The mITT population excluding all participants with protocol violations that were determined by the protocol deviation adjudication committee to be exclusionary from the per-protocol population.

6.5 Injection (Step 2) Efficacy Population

The mITT population who receive at least one injection and are uninfected at the time of the first injection, and had at least one follow-up visit with HIV status determined after the first injection.

6.6 Safety Population (Primary Safety)

All ITT participants who received any oral or injectable product.

6.7 Injection (Step 2) Safety Population

All Safety Population participants who receive at least one injection

6.8 Longitudinal Pharmacokinetic CAB Concentration Population

A longitudinal evaluation of CAB PK in the CAB arm will be conducted in 150 participants, excluding any participant who has had any positive HIV testing at any visit (exclude HIV final status= "positive" or "additional testing needed"). Participant selection should be limited to participants who have reached week 57 before 3.1.2020 and have no late injections beyond 2 weeks of target date prior to week 57. The window for late injections is defined as 2 weeks after target date. Selected participants should have no missed injections (including regular injection visits, as well as no missing injection). These 150 participants present the following distribution across sites:

- 40% cisgender women <25 years of age at enrollment
- 40% cisgender women between 25 and 30 years of age at enrollment
- 20% cisgender women ≥ 30 years of age at enrollment

This population will be used for the CAB concentration listing and to enhance the population represented in the global modelling of CAB-LA PK.

6.9 TDF-FTC Adherence Population

The TDF-FTC adherence population consists of approximately 400 participants in the TDF/FTC arm who were randomly selected for assessment of adherence to oral pill taking. To achieve balance across geographic regions and balance over time, selection of the Adherence Population was stratified by study sites or regions and was selected throughout the enrollment period by randomly choosing approximately 25% of all new enrollees in the TDF/FTC arm each quarter. Plasma and DBS samples for selected follow-up visits are tested for TFV and FTC throughout the follow-up period.

6.10 Contraceptive Substudy Population

All participants who are invited and agree to participate in the Contraception sub study.

6.11 Pregnancy Population

All participants who have a positive pregnancy test result while on treatment.

6.12 Confirmed Pregnancy Population

All participants who have a positive pregnancy test result while on treatment that is followed by a positive confirmatory test result or confirmation by another method. (Ultrasound, full- or pre-term live birth.).

6.13 Declined contraception population

All participants who decline to use long-acting contraception at any point during step 1 or 2 after LOA 3 to protocol v1.0 was implemented.

6.14 Seroconverter Population

All ITT participants who, during the course of the trial, are either identified to have been HIV infected prior to randomization, or who acquired HIV infection during trial the primary analysis follow-up period.

6.15 Primary Seroconverter Population

All ITT participants who are HIV-uninfected at randomization and acquire HIV infection during the primary analysis follow-up period.

6.16 Infected at Enrollment Population

All ITT participants who were subsequently determined to be HIV-infected at randomization.

6.17 Subgroups

Important participant subgroups are

- Age: <25 vs ≥25 years old
- HSV-2 status at baseline*
- Method of contraception during follow-up**
- BMI </≥ 30 kg/m²

* Note: HSV-2 status at baseline will not be available for the Clinical Study Report

**Note: Contraceptive method will be modeled as a time-dependent covariate. This analysis will not be presented in the Clinical Study Report.

7. STATISTICAL ANALYSES

7.1 General Analysis Considerations

Any deviations from the original statistical plan will be described in the final report.

7.1.1. Summary Statistics

Summary statistics for categorical variables will include frequency and percentage. For continuous variables, number of subjects with non-missing value (n), mean, median, standard deviation (SD), minimum (min), and maximum (max) will be reported. For change from baseline calculations, baseline is defined as Day 0/Randomization visit. If a baseline value is not available from the randomization visit, for certain parameters the baseline value may be obtained from the enrollment visit.

7.1.2. Visit Windows

All data will be tabulated per the evaluation visit as recorded on the CRF even if the assessment is outside of the visit window. Visit windows are displayed in Appendix A. In

data listings, the relative day, of all dates will be presented e.g. from date of randomization, or date of last injection.

7.2. Interim Analyses

An independent Data Safety and Monitoring Board (DSMB) will meet approximately every six months throughout the study. The DSMB will do the following:

- I. Periodically review and evaluate the accumulated study data for participant safety, study conduct and progress and treatment efficacy.
- II. Make recommendations to the study investigators and DAIDS concerning the continuation, modification, or termination of the trial.

The DSMB will consider study-specific data as well as relevant background knowledge about the disease, test agents and patient population under study.

Analyses presented to the DSMB will include information on accrual and retention, comparability of the CAB LA and TDF/FTC arms at enrollment, and adherence, adverse events and HIV incidence by arm. The DSMB may also ask for additional analyses they deem relevant.

Below we provide guidelines for monitoring HPTN 084 for:

Study progress and validity	Futility monitoring
Efficacy monitoring	Safety monitoring

7.2.1. **Study progress and validity**

At each meeting the DSMB will review data on enrollment (as necessary), retention of participants, delivery of the intervention and all protocol violations. The DSMB will also compare the (pooled) observed HIV incidence rate to the (pooled) expected HIV incidence rate to determine if the HIV incidence rate among study participants is similar to the incidence rate assumed in the study design. The DSMB will recommend any changes to the study design they believe are needed to ensure study validity.

Expected event rate: 2.07/100 person-years in the TDF/FTC arm and 1.12/100 person-years in the treatment arm; 1.6/100 person-years overall. Note: In general, person-years are calculated as the sum of the total follow-up time for all participants included in the analysis, using the censoring rules as described for each individual analysis. Start and end dates and censoring rules for the primary analysis are defined in Appendix B.

7.2.2. **Efficacy monitoring**

This analysis determines the Sample Size of the study, therefore kindly refer to section 4.1.1 of the SAP.

7.2.3. **Futility monitoring**

The DSMB may determine that the trial is operationally futile if enrollment rates are substantially lower than expected (such that accrual cannot be completed in a timely manner) or retention is so poor that results of the trial would lack validity.

However, other aspects of the trial which are typically evaluated for futility (low HIV event rate, lack of difference between the arms) must be evaluated in the context of participant adherence to the study drugs. While high adherence is expected in the CAB LA arm, high rates of adherence in the TDF/FTC arm would likely lead to a lower than expected HIV event rates and little difference in HIV incidence between the arms. Such a result should

not lead to a futility finding. While HPTN 084 is designed as a superiority trial, a contingency plan for a supplementary non-inferiority analysis in the case of high TDF/FTC adherence has been developed. This analysis is described in detail in the **protocol HPTN 084 v2.0, in the section 7.8.4 entitled “Considerations for a Supportive Analysis if Adherence to TDF/FTC is Higher than Expected”**. To aid in this evaluation, detailed information on adherence to the study drugs, by arm, will be provided to the DSMB.

7.2.4. Safety monitoring

At each 6 months visit the DSMB will be provided with a list of all adverse events, by grade and broken down by study arm. The relationship of the event to the treatment (definitely related, probably related, possibly related, not related) will also be described. In the closed report at each visit the study team will provide the DSMB with a (nominal) p-value comparing the rate of grade 3 or higher adverse events in the two study arms. However, no formal stopping rule for safety monitoring is proposed so as to give the DSMB maximum flexibility in balancing the risks and benefits of the treatments.

7.3. Endpoint Analyses

7.3.1. Primary Efficacy Analysis

Objective: To evaluate the relative efficacy of oral CAB/CAB LA (oral run-in and injections, Steps 1 and 2) vs. daily oral TDF/FTC for HIV prevention

Population: mITT population

Outcome: Time of HIV-1 Infection (censored at end of step 2)

Time from randomization to HIV-1 infection (see definition in appendix B) will be the primary endpoint for this analysis. HIV-1 status will be assumed to be negative for any missing visits prior to the first positive HIV-1 test (unless defined differently by the endpoints committee prior to analysis). Person-time and HIV events will be included in this analysis based on each individual's scheduled participation in steps 1 and 2, as determined at randomization. Specifically, individuals who refuse/discontinue injections, pills or both, or who receive open-label study product (e.g. due to pregnancy) will be included in this analysis in their original randomization arm for the duration of their originally scheduled participation in Steps 1 and 2. Individuals who drop out of the study and refuse further testing prior to completion of follow-up and individuals who die prior to completion of follow-up will be treated as uninformatively censored as of their last valid HIV test.

Descriptive analysis: The number of participants, number of infections and cumulative person-years will be presented by arm and overall. The HIV incidence rate will be calculated as the total number of participants with confirmed incident HIV infection during steps 1 and 2 of study follow-up, divided by the person-years accumulated in each arm. Corresponding 95% CIs will be calculated based on Poisson assumptions. Cumulative incidence over follow-up for each arm will be computed using product limit estimates and plotted with 95% CIs.

Statistical Analysis: The primary “intent-to-treat” analysis will be based on a Cox model using time to HIV detection (as defined above) as the outcome. The model will include an indicator for intervention arm (TDF/FTC = 0; CAB LA = 1) and be stratified by site. Consistent with an intent-to-treat analysis, women will be included in the analysis in their original randomization group whenever endpoint information is available within the prespecified timeframe of follow-up. Participants who remain HIV-1 uninfected will be censored at their last negative HIV test. A formal statistical comparison between study arms will be performed using a Wald test of the hypothesis

$H_0: HR = 1$

$H_a: HR < 1$

A two-sided test with $\alpha = .05$ will be used. The estimated hazard ratio (HR), a 95% confidence interval, and a p-value for the hypothesis shown above will be given. If the number of events is small (<40) then the p-value will be confirmed using a permutation test based on 100,000 random permutations of the treatment assignments; if there is a meaningful difference between the permutation and asymptotic procedures, the permutation p-value will be used. Treatment efficacy will be estimated as $TE = 1 - HR$.

If the trial is stopped early (prior to the final analysis), then the primary results will be reported as a bias-adjusted mean HR and design-adjusted 95% CI (Emerson S and Fleming TR. Symmetric Group Sequential Test Designs. Biometrics 45:904-923, 1989).

7.3.1.1. Sensitivity of the Primary mITT results:

The following sensitivity analyses are proposed to aid in interpretation of the primary ITT results :

- A per-protocol analysis, excluding participants with exclusionary protocol violations, will be conducted if the number of participants affected exceeds 2% of the enrolled population.
Population: Per Protocol
This analysis will be identical to the mITT analysis, except that for partial exclusions, the follow-up will be censored on the last non-missing HIV test result date on or prior to the date of the deviation that resulted in exclusion from the per-protocol population.
- A supportive analysis will be presented using the OBSP censoring in the Injection (Step 2) Efficacy population (see OBSP censoring definitions in appendix B), where study follow-up is censored when a participant does not receive blinded injection study product on time.
- In this active control trial, the superiority of CAB LA is predicated in part on incomplete adherence to daily oral TDF/FTC. Specifically, we assume that overall adherence to TDF/FTC will be less than 50%. If adherence to TDF/FTC is high (greater than 50%) then, as described in section 7.8.4 of the protocol, a supportive non-inferiority (NI) analysis of CAB LA versus TDF/FTC may provide valuable information for interpreting the trial results.

Adherence in the TDF/FTC arm will be measured during the trial in an adherence assessment cohort (see below). If adherence in the TDF/FTC arm is greater than 50% then an NI margin will be computed as described in section 7.8.4.3 of the protocol. The primary analysis described above will then be repeated using the recomputed NI margin instead of 1.0 in the null hypothesis. This supportive NI analysis will not replace the pre-specified primary analysis or change the conclusion of the primary analysis but may provide valuable insight with respect to determining the prevention effectiveness of CAB LA in women.

7.3.1.2. Adherence Assessment Cohort:

A sample of 400 participants in the active TDF/FTC arm will be selected at random during the enrollment period and available plasma samples from selected visits (Step 1: week 4; Step 2: weeks 9, 17, 33, 57, 81, 105, 129, 153, 177) will be assessed for the presence of TDF/FTC and associated metabolites. The TDF/FTC adherence-assessment cohort will be stratified by site or

region and will be selected to ensure uniform representation over the course of the enrollment period. Average, study-wide adherence will be computed as the overall proportion of all tested samples where study drug (TDF/FTC) is detected (i.e. concentrations greater than approximately 0.31 ng/ml).

7.3.2. Primary Safety Analysis

Objective: To evaluate the relative safety of oral CAB/CAB LA (oral run-in and injections, Steps 1 and 2) vs. daily oral TDF/FTC for HIV prevention (Steps 1 and 2).

Population: Safety population

The primary pre-specified safety comparison between arms will be based on all AEs with maximum grade ≥ 2 . In general, formal statistical comparison of individual types of safety events across arms is not planned since interpretation of differences must rely heavily upon clinical judgment. Where formal statistical testing is considered necessary to guide judgement of observed differences for any single event or collection of events, differences in events grade 2 and higher will be assessed. The event rate and 95% robust confidence interval for each treatment arm will be calculated using a log-linear model for counts using $\log(\text{person-time})$ as an offset, treatment as a covariate, assuming a Poisson distribution and using a robust variance (i.e. generalized estimating equations). Event rates will be compared between treatment arms using this same model. Event rates will be compared between treatment arms using this same model and a type I error rate of $\alpha=0.05$ is prespecified. Very rare events will be compared using an exact Poisson test. Additional supporting descriptive safety analyses will be presented.

For Safety OBSP analysis, adverse event follow-up will be censored after the last injection before a termination or a permanent product discontinuation regardless of any delays in injections. If termination or a permanent product discontinuation occurs prior to their first injection, the participant's follow-up time will be censored at the termination or permanent product discontinuation date +1. If termination or a permanent product discontinuation occurs after the first injection, the participants follow-up time will be censored up through 6 weeks after the Week 5 injection, if that is the last injection, or 10 weeks after the last injection for subsequent injections.

7.3.3. Adverse events in Step 1

Objective: To compare the safety of oral CAB vs. oral TDF/FTC in Step 1

Population: Primary Safety

Censoring: AEs are censored at 120 days after randomization for participants who never received an injection; at 1 day after the date of termination or investigational product discontinuation; at date of first injection, whichever occurs first. Reported AEs will be excluded if they occur after the censoring time.

Descriptive and Statistical Analysis as above

7.3.4. Adverse events Step 2 only OBSP

Objective: To compare the safety of CAB LA vs. oral TDF/FTC proximal to blinded study drug administration

Population: Step 2 safety

Censoring: AEs are censored using ON BLINDED STUDY PRODUCT (OBSP) censoring.

Details of the OBSP censoring definitions are given in Appendix B. Reported AEs will be excluded if they occur before the first injection or after the OBSP censoring time.

Descriptive and Statistical Analysis will be performed as above

7.3.5. Laboratory evaluations

Population: Primary Safety

Descriptive: Laboratory findings for CBC, chemistries (urea, creatinine, CPK, glucose, calcium, phosphorous, amylase, lipase), liver function (AST, ALT, total bilirubin, alkaline phosphatase) and fasting lipid profile (total cholesterol, triglycerides, HDL, LDL) will be reported by grade, as defined in the “Division of AIDS (DAIDS) Table for Grading the Severity of Adult and Pediatric Adverse Events”. The proportion of laboratory findings qualifying as Grade 2 or above will be presented. Laboratory values will be reported by arm as median, 1st and 3rd quartiles, min and max, and (at follow-up visits) as median, 1st and 3rd quartile change from baseline (see below). Summaries involving values and changes from baseline at scheduled visit time points will consider values from scheduled visits only, and in the case of missing values, results will not be carried forward from interim (unscheduled) visits. For summaries and analyses involving changes in laboratory values over time or maximum change from baseline (e.g. shift tables), values from interim (unscheduled) visits will also be considered.

Formal comparisons of laboratory values will be conducted as follows:

- Absolute values: A mixed-effects model with participant as the random effect and site and arm as fixed effects will be used to assess overall differences in each laboratory outcome by arm. Outcomes with highly skewed distributions will be log-transformed. A continuous time (elapsed study time) term and an interaction of time with study arm will be added to model time trends in laboratory values that may emerge with continuous exposure to study drug. Only post baseline visits will be included.
- Change from baseline. A fixed effect model with site and arm as covariates will be used to assess overall change from baseline for each laboratory outcome by arm. A continuous time (elapsed study time) term and an interaction of time with study arm will be added to model differences in laboratory values that may emerge with continuous exposure to study drug.

Note: Formal comparisons of laboratory values as described above will not be included in the Clinical Study Report.

7.3.6. Secondary Analyses

- a. **Objective:** To compare HIV incidence among participants receiving oral CAB/CAB LA vs. daily oral TDF/FTC (Steps 1, 2 and 3).

Population: mITT population

Outcome: Time of HIV-1 Infection

This analysis will be similar to the primary analysis of HIV incidence but will include the entire follow-up period (steps 1, 2 and 3) for each participant. The analysis will be based on a Cox model using time to HIV detection (as defined in the primary analysis) as the outcome. The model will include an indicator for intervention arm (TDF/FTC = 0; CAB LA = 1) and be stratified by site. Participants who remain HIV-1 uninfected will be censored at their last negative HIV test. A formal statistical comparison between study arms will be performed using a Wald test of the hypothesis

$H_0: HR = 1$

$H_a: HR < 1$

A one-sided test with $\alpha = .025$ will be used. The estimated hazard ratio (HR), a 95% confidence interval, and a p-value for the hypothesis shown above will be given.

- b. **Objective:** To evaluate relative efficacy of oral CAB/CAB LA vs. oral TDF/FTC in subgroups defined by the baseline factors of age, herpes simplex virus-2 (HSV-2) serostatus, and body mass index (BMI).

Population: mITT population

Outcome: Time of HIV-1 Infection

These analyses will be similar to the primary analysis of HIV incidence and will include data from follow-up steps 1 and 2 for each participant. The number of events, person-years and incidence will be calculated for each arm within each subgroup.

Subgroups will be defined as follows:

Age (<25 vs ≥ 25)

HSV-2(positive or negative)

BMI (<30.0 vs ≥ 30.0)

Contraceptive method during treatment (DMPA, Net_EN, Implants, Other)

Age, HSV-2 and BMI will be based on baseline status while contraception will be treated as a time-varying covariate.

The analyses of efficacy will be based on a Cox model using time to HIV detection (as defined in the primary analysis) as the outcome. Separate models will be fit for each of the three baseline covariates of interest. Each model will be stratified by site and will include an indicator for intervention arm (TDF/FTC = 0; CAB LA = 1), the covariate of interest and their interaction.

A formal statistical comparison of efficacy between the subgroups will be performed using a Wald test of the hypothesis

$H_0: \text{interaction term} = 0$

$H_a: \text{interaction term} \neq 0$

A two-sided test with $\alpha = .05$ will be used.

The efficacy (relative hazard) of CAB LA versus TDF/FTC and 95% CI will be calculated for each subgroup using standard methods for linear combinations of the Cox model coefficients.

- c. **Objective:** To describe and model the relationship between HIV incidence and drug concentration, within each arm.

Population: PK analysis population

Outcome: Time of HIV-1 Infection

These analyses will include data from follow-up steps 1 and 2 for each participant in the PK analysis population. Separate analyses will be conducted for each arm. The analyses will be based on a logistic regression of all available intervals with drug concentration/PK parameter data. The outcome will be HIV infection (yes/no) as measured at the end of the interval. For each arm, the drug concentration/ PK parameter of interest in each interval will be determined in conjunction with colleagues from the central lab and ViiV (see separate document with more details on the PK study design). A plot of the model predicted relative hazard of HIV infection (y-axis) versus drug concentration (x-axis) with a 95% confidence interval will be generated from the final model.

- d. **Objective:** To describe the distribution and correlates of drug concentration, within each arm.

Population: PK analysis population

Endpoint: Drug concentration

These analyses will include data from follow-up steps 1 and 2 for each participant. Separate analyses will be conducted for each arm.

Descriptive analyses: A histogram of all measured drug concentrations/PK parameter estimates will be generated. Boxplots of these outcomes will be generated both overall, over time and for key subgroups (age, HSV-2 serostatus, BMI, reported/document adherence).

Statistical Analyses: Multiple linear regression analyses with correction for correlated data and robust variances (i.e. generalized estimating equations) will be used to relate log of the drug concentration to age, HSV-2 serostatus, BMI and any other covariates of interest. Reported (oral pills) or documented (injection) level of adherence will be included in the models. The coefficient of each covariate and a 95% confidence interval will be provided.

- e. **Objective:** To compare the acceptability of and preferences for CAB LA vs. oral TDF/FTC.

Population: mITT population

Outcomes: Two primary outcome variables will be used:

- Response to CASI question (step 3, day 0, visit 33.0) “If you wanted to protect yourself from getting HIV, which of the following product(s) would you prefer to use?”
- Response to (Likert-scale) CASI questions (step 3, day 0, visit 33.0) i) “How often do you find it inconvenient or difficult to take your oral study medication (i.e. the tablets) as recommended?” ii) “How often do you find it inconvenient or difficult to receive your injection as recommended?”

Descriptive analysis: Responses to these questions will be summarized by arm for the CSR.

Statistical Analysis: The CSR will only display descriptive summaries as described above. Subsequent analyses will be conducted on data collected at the end of step 2 (visit 33.0). Chi-square tests and Fisher-exact tests will be used as appropriate, with Likert scales converted to Binary by distributions within the response levels (e.g. at the median response). Inference for differences by arm will be analyzed using logistic regression models. Each model will contain two terms:

- Country
- Randomized study arm.

7.3.7. Tertiary Analyses

- a. **Objective:** To estimate sexual risk behaviors, as measured by self-report and rates of incident sexually transmitted infections (STIs).

Population: mITT population

Outcomes:

- i. Sexual risk
 - Any condomless sex with an HIV-infected or HIV-unknown status partner
 - >1 sexual partners in past 1 months
- ii. Incident STIs (measured at weeks 33, 57, 81, 105, 129, 153 and 177 during Step 2, and weeks 24 and 48 during Step 3)
 - Time to incident STI (any of Gonorrhea, Chlamydia, Syphilis)

Analysis: Data from steps 1, 2 and 3 will be used unless indicated otherwise.

The proportion of participants reporting any high-risk behavior will be assessed at each visit. Mixed effects logistic regression will be used to estimate prevalence and 95% confidence intervals for each behavior. Time trends during follow-up will be estimated using mixed effects logistic regression. Linear and non-linear models (with respect to time) will be explored.

Rates of incident STI's will be computed as number of new STI's divided by total person-years. Poisson regression with a robust variance will be used to model the number of incidents STI for each woman. Person-time will be used as an offset. Incidence and a 95% CI will be reported overall and by arm.

- b. **Objective:** To compare Grade >2 adverse event (AE) rates in women with baseline BMI </≥ 25 kg/m², within each study arm.

Population: Primary Safety Population

Outcome: Counts of grade 2 or higher AE

For each arm, tables will summarize the number of grade 2 or higher AEs by MedDRA System Organ Class and BMI. Participant with multiple AEs within a category will be counted once under the maximum severity. The overall grade 2 or higher event rate and a 95% confidence interval will be calculated for those with BMI < and > 30 kg/m² at baseline based on a log-linear model for counts using log(person-time) as an offset, assuming a Poisson distribution, and using a robust variance (i.e. generalized estimating equations). Event rates will be compared between those with BMI < and ≥30 kg/m² at baseline using a similar model.

- c. **Objective:** To compare differences in weight gain, by arm.

Population: Primary Safety Population

Outcome: Weight (kg)

Weight gain will be computed as change from baseline (enrollment) at each follow-up visit. We will plot mean weight gain over follow-up by treatment arm. A linear mixed model with random person and time effects will be fit with (i) continuous time and (ii) categorical intervals for time in study, treatment and time by treatment interaction to evaluate the effect of CAB LA on weight gain. The interaction between time and treatment will quantify the difference in weight gain by arm. A second analysis that removes women when they become pregnant will be conducted.

- d. **Objective:** To compare pregnancy incidence and outcomes between arms.

Population: mITT Population and PREGNANCY Subgroup Population

Outcome: New pregnancies and pregnancy outcomes

Data from steps 1 and 2 (PRIMARY mITT dataset) will be used. Tables will summarize the number of pregnancies and pregnancy rate by arm. The pregnancy rate and 95% robust confidence interval for each treatment arm will be calculated using a log-linear model for counts using log(person-time) (excluding time spent pregnant) as an offset, assuming a Poisson distribution, and using a robust variance (i.e. generalized estimating equations).

Outcomes of all detected pregnancies (PREGNANCY dataset) and all confirmed pregnancies (CONFIRMED PREGNANCY dataset), including information of drug exposure during the conception period and pregnancy period, will be tabulated by arm.

- e. **Objective:** To evaluate rates of HIV drug resistance among participants who acquire HIV infection during the study among participants receiving oral CAB/CAB LA vs. oral TDF/FTC.

Population: Primary SEROCONVERTER Population

Data from steps 1, 2 and 3 will be included. The number of cases of drug resistance will be summarized by arm and step. All cases of drug resistance among incident HIV infections during the study will be described.

- f. **Objective:** To determine plasma concentrations of medroxyprogesterone (DMPA) or norethisterone (NET-EN) or etonorgestrel (from implants) when co-administered with CAB LA.

Population: CONTRACEPTION Subgroup Population

Outcome: Plasma concentration of medroxyprogesterone or norethisterone.

Boxplots of plasma concentrations of DMPA/NET_EN/etonorgestrel will be generated by arm. For women in the CAB-LA arm, a plot of plasma concentrations versus time since last injection will be plotted and a smooth line will be fit through the data. Proportion of women with plasma concentrations below 0.1 ng/mL (DMPA) or TBD (NET_EN) or 90pg/mL (etonorgestrel) will be reported and compared (separately, for women using each drug) between arms using a Fisher's exact test or a permutation test if multiple observations are available on each woman.

In addition, we will compare mean contraceptive drug levels (and/or other PK parameters) between the CAB LA and TDF/FTC arms for each contraceptive subgroup using a t-test (or a t-test with robust variance if multiple observations on each woman are available).

8. MISSING DATA AND IMPUTATIONS

Data may be missing due to participant dropout, participant nonresponse, or due to failures in data collection. In general, the last category (failures in data collection) is quite rare in HPTN trials. Further, it is common to assume that such data are missing completely at random (MCAR) (Rubin DB. *Biometrika* 1976; 63:581–592) and so will not bias results. However, participant dropout and/or nonresponse are unlikely to be MCAR and so attention must be paid to the potential for bias if there are high levels of missing data for these reasons. Here we discuss the approaches we will take for analyses in the presence of missing data.

For the primary endpoints, HIV infections and safety events, and other biological endpoints the primary cause of missing data is participant dropout. Historically, HPTN trials have had good retention of participants and we expect that to be the case for this trial. Therefore, analyses will be conducted assuming uninformative censoring. If loss-to-follow-up is low and similar between the arms we will not conduct any additional sensitivity analyses. However, if loss-to-follow-up is greater than 20% or meaningfully different between arms (>5%-points), then we will investigate the sensitivity of the results to assumptions about the missing data. Specifically, we will use inverse probability-of-censoring weights to adjust for loss-to-follow-up (Cain LE. *Stat Med* 2009; 28:1725-1738) and compare the adjusted treatment effect to the unadjusted treatment effect.

Behavioral and self-reported endpoints (e.g. acceptability) may be subject to participant nonresponse. The analyses described below are based on a complete-case approach. However, if non-response is high (>15%) or differential between arms (> 5%-points) then we will perform sensitivity analyses using multiple imputation. Baseline data will be used to develop the imputation model and standard errors will be adjusted using Rubin's method (Rubin DB. *Multiple Imputation for Nonresponse in Surveys*, Wiley, New York, 1987). Both complete case and multiple imputation results will be reported.

8.1. Missing Start and Stop Dates for Prior and Concomitant Medication, and Medical History

The Assumption is to consider the most conservative judgment when imputing.

Start date:

When imputing a start date ensure that the new imputed date is sensible i.e. is prior to the end date and the general rules are:

- i. If start date is completely missing, start date will not be imputed
- ii. If year and month are present and day is missing, set day to the 1st day of month.
- iii. If year and day are present and month is missing, then set month to January.
- iv. If year is present and month and day are missing, set month and day to January 1st.

Stop date:

- i. If end date is completely missing, end date will not be imputed
- ii. If year and month are present and day is missing, set day to the last day of month.
- iii. If year and day are present and month is missing, set month to December

- iv. If year is present and month and day are missing, set month and day to December 31st.

8.2. Missing Start and Stop Dates for Adverse Events

The Assumption is to consider the most conservative judgment when imputing.

Start date:

1. If start date is completely missing, start date is set to date of first dose.
2. If (year is present, and month and day are missing) or (year and day are present, and month is missing):
 - a. If year = year of first dose, then set month and day to month and day of first dose.
 - b. If year < year of first dose, then set month and day to December 31st.
 - c. If year > year of first dose, then set month and day to January 1st.
3. If month and year are present and day is missing:
 - a. If year = year of first dose and
 1. If month = month of first dose, then set day to day of first dose date.
 2. If month < month of first dose, then set day to last day of month.
 3. If month > month of first dose, then set day to 1st day of month.
 - b. If year < year of first dose, then set day to last day of month.
 - c. If year > year of first dose, then set day to 1st day of month.

Stop date:

If the outcome of the AE was ongoing or unknown, then the rules outlined below will not be applied.

1. If stop date is completely missing, stop date is set to date of study discontinuation.
2. If (year is present, and month and day are missing) or (year and day are present, and month is missing):
 - a. If year = year of study discontinuation, then set month and day to month and day of study discontinuation.
 - b. If year < year of study discontinuation, then set month and day to December 31st.
 - c. If year > year of study discontinuation, then set month and day to December 31st.
3. If month and year are present and day is missing:
 - a. If year = year of study discontinuation and
 - i. If month = month of study discontinuation, then set day to day of study discontinuation date
 - ii. If month < month of study discontinuation, then set day to last day of month.
 - iii. If month > month of study discontinuation, then set day to last day of month.
 - b. If year < year of study discontinuation, then set day to last day of month.
 - c. If year > year of study discontinuation, then set day to last day of month.

9. MONITORING REPORTS

9.1. Performance Metric, SMC & DSMB Reports

This section describes the main tables and reports. A subset of these will be presented to the DSMB at each meeting. When results are presented by study arm, the arms will be labeled Arm A and Arm B. A separate document attached to the DSMB report will unblind the arms (i.e., TDF/FTC or CAB). The study accrual table will report on all enrolled women; subsequent tables will report on women who are HIV-uninfected at enrollment only.

Real-time reports will be available to the protocol team. The content of these reports will be developed by the protocol team and will generally include the following information:

- **Study Accrual**
The number of individuals enrolled and randomized per study month will be presented in a table overall and (1) by site and (2) by study arm. No formal statistical testing will be performed.
- **Baseline Demographic and Clinical Characteristics of Young women**
A table that includes baseline demographic and clinical characteristics (e.g. age, employment, marital status, education, race, BMI, HSV-2 status, etc.) will be presented. Summary statistics appropriate for the measurement scale will be used to describe the distribution of these variables. These summary statistics will be presented in tables overall, by site and by study arm. No formal statistical testing will be performed.
- **Baseline Behavioral Characteristics**
A set of tables will describe baseline behavioral characteristics for participants. Measures included in these tables will include age of sexual debut, and the number of partners in lifetime and the last 3 months, contraceptive method, and STI. Summary statistics appropriate for the measurement scale will be used to describe the distribution of these variables. These summary statistics will be presented in tables overall, by site and by study arm. No formal statistical testing will be performed.
- **Retention**
A set of tables will describe retention in follow-up for participants at each scheduled study visit. Number of women enrolled, number of women expected at each visit, number of women attending the visit early, late and on time, and percent retained will be presented. A separate retention report will be generated for women who have moved to alternative visit schedules (e.g. pregnancy, open-label TDF/FTC, annual testing); these women's visits will not be included in the standard retention table after they move to an alternative visit schedule. Retention will be presented overall and by site in the open report and by study arm in the closed report. No formal statistical testing will be performed.
- **Fidelity of Intervention**
A set of tables will describe the proportion of injections provided and the proportion of pill bottles distributed at each visit, among those retained at the visit. These tables will be presented overall and by site in the open report and by study arm in the closed report. No formal statistical testing will be performed.
- **Safety Reports (Closed Reports)**
The number and percentage of participants experiencing local reactions to the injections will be tabulated by severity and treatment arm. For a given local reaction type, each participant's

reaction will be counted once under the maximum severity for all injection visits. In addition to the individual reaction types, the maximum severity of local pain or tenderness, induration or erythema, and of systemic symptoms will be calculated. AEs will be summarized using MedDRA System Organ Class and preferred terms. Tables will show by treatment arm the number and percentage of participants experiencing an AE within a System Organ Class or within preferred term category by severity. For the calculations in these tables, a participant with multiple AEs within a category will be counted once under the maximum severity.

A listing of SAEs reported to the DAIDS Regulatory Support Center (RSC) Safety Office will provide details of the events including severity, time between onset and last dosing, and cumulative number of doses received.

Box plots of local laboratory values will be generated for baseline values and for values measured during the study by treatment arm and visit. Each box plot will show the first quartile, the median, and the third quartile. Outliers will also be plotted. If appropriate, horizontal lines representing boundaries for abnormal values will be plotted. The number (percentage) of participants with local laboratory values recorded as meeting Grade 3 AE criteria or above as specified in the DAIDS AE Grading Table will be tabulated by treatment arm. Reportable clinical laboratory abnormalities without an associated clinical diagnosis will also be included in the tabulation of AEs described above.

- **Efficacy Reports**

The efficacy reports will be shared with the DSMB committee when required as per the analysis methodology mentioned in section 7.2.2 and 7.3.1 of the SAP ()above.

9.2. Clinical Study Report

The tables described below will be included in the CSR. Additional tables and analyses may be included, at the discretion of the protocol team or sponsor, e.g., depending on the availability of additional laboratory evaluations. All reports and analyses will be presented by study arm.

Subjects will be pooled across all sites. Data will be presented by treatment group and overall. Categorical variables will be summarized by frequency distributions (number and percentages of subjects) and continuous variables will be summarized by descriptive statistics (mean, standard deviation, etc.).

Adverse events will be summarized by treatment group using the Medical Dictionary for Regulatory Activities (MedDRA) system organ class and preferred term. Tabulations will be produced for all treatment-emergent AEs (TEAEs) defined as AEs with an onset date on or after the start of treatment, treatment-related AEs (those considered by the Investigator as at least possibly drug related), SAEs, discontinuations due to AEs, and AEs by severity grade. By-subject listings will be provided for any deaths, SAEs, and AEs leading to discontinuation.

Descriptive statistics for each treatment group will be provided for clinical laboratory values (e.g., hematology, serum chemistry, and urinalysis) and vital signs data, presented as both actual values and changes from baseline relative to each on-study evaluation. Abnormal clinical laboratory values will be listed. All data will be provided in by-subject listings.

Displays of the cumulative incidence of events such as HIV infection, or treatment discontinuation, will include at-risk tables as follows: Numbers of subjects having had the event of interest, and at risk to have had the event will be shown at selected weekly timepoints under the horizontal axis. The number at risk will be computed as the number of subjects with event times greater than or equal to the given timepoint

- **Subjects Disposition**

The subject disposition summary will include the number screened, the number of screen failures, the number enrolled, the number in each patient population for analysis, the number who completed the study, the number who discontinued the study and reason for discontinuation from the study.

Disposition data will be summarized by treatment and overall. A by-subject data listing of study completion information including the reason for study discontinuation will be presented. A by-subject listing of inclusion/exclusion criteria not met will also be presented.

- **Protocol Deviations**

A summary of all protocol deviations on ITT population by type will be generated. Protocol deviation data will be summarized by treatment and overall. A by-subject data listing of protocol deviations will also be presented. All protocol deviations will be presented in a data listing.

- **Demographic and Baseline Characteristics**

Demographic and baseline characteristic data summarization will be performed to descriptively assess the comparability of treatment groups. Data to be tabulated will include age, race, ethnicity, height, weight, and BMI, as well as other baseline characteristics.

- **Exposure and Adherence**

Drug Exposure and adherence will be calculated and will be presented in a tabulation form and by listing.

- **Medical History**

A by subject listing of verbatim medical history will be presented.

- **Concomitant Medication**

Concomitant medications will be coded using the World Health Organization (WHO) dictionary. Results will be tabulated by anatomic therapeutic class (ATC) and preferred term. This data will be summarized by treatment group. Previous and concomitant medications will be presented in a data listing.

- **Vital Signs and Physical Examination**

Vital sign measurements will be presented for each subject in a data listing. Systolic blood pressure, diastolic blood pressure, heart rate, and weight will be summarized as actual value and change from baseline by visit. All physical examination findings will be presented in a data listing.

- **Laboratory Data**

Clinical laboratory values will generally be expressed in SI units .

The actual value and change from screening will be summarized for each clinical chemistry, hematology, lipid panel, HbA1C and urinalysis parameters and by each visit. In the event of repeat values, the last non-missing value per visit will be used.

All laboratory data will be also provided in data listings. Values outside of the lab parameter's normal range will be flagged as high, low, or abnormal based on the range of the test. Laboratory results will be summarized by scheduled study visit. Summary calculations such as maximum post-baseline level will include data from interim (unscheduled visits). Formal comparisons of laboratory findings between arms will not be included in the CSR.

- **Adverse Events**

The reporting of safety data is descriptive and will include all subjects who receive at least one dose of investigational product. AEs will be summarized based on the frequency of AEs and their severity for all treated subjects. All AEs will be coded using Medical Dictionary for Regulatory Activities (MedDRA) and summarized by treatment group. Data will be summarized using preferred term and primary system organ class.

If a subject experiences multiple events that map to a single preferred term, the greatest severity will be assigned to the preferred term for the appropriate summaries.

Summaries of AEs will include any AEs reported beginning with the first dose of investigational product on Day 1. The occurrence of adverse events will be summarized by treatment group using preferred terms, system organ classifications, and severity. Separate summaries of serious adverse events, adverse events related to investigational product, and events leading to the discontinuation of investigational product will be generated. All adverse events reported will be listed for individual subjects showing both verbatim and preferred terms.

- **Efficacy and Efficacy Subgroups**

In the Clinical Study Report, primary efficacy analysis will be presented as described in Section 7.3.1. For subgroup analyses, the age and BMI subgroup analyses will be presented in the Clinical Study Report, but analyses considering contraceptive use and HSV-2 status at baseline will not be presented in the Clinical Study Report.

10. Summary of changes

10.1 Version 3.0 was prepared during the planning process for the Clinical Study Report. The blinded statistician continued to not have access to individual level data.

10.1.1. Analysis Populations

- Populations were adjusted, renamed, or clarified: mITT; Per Protocol; Seroconverter; Contraceptive Substudy; Pregnancy.
- Populations were added: Randomized; ITT; Primary Safety; Step 2 Safety; Step 2 Efficacy; Longitudinal Pharmacokinetic Cabotegravir; TDF/FTC Adherence Population; Confirmed Pregnancy; Seroconverter Population; Primary Seroconverters; Declined Contraception; Infected at Enrollment
- The Version 2.0 "Per-Protocol" analysis dataset was redefined and renamed as On Blinded Study Product (OBSP) censoring. The new "Per-Protocol" population was

defined as being based upon deviation adjudication. The adjudication process was described.

- The age and BMI subgroups were modified with adjusted cut-points.

10.1.2. Analysis Details and Clarifications

- Clarifications and additional details were added throughout the Statistical Analysis section.
- A section was added to describe missing data handling and imputations.
- Appendices were added to provide detail for visit windowing, efficacy and safety analysis censoring details and OBSP specifications, a listing of study sites, a listing of adverse events of special interest (AESI). Key variable definitions, which were based on data fields found in the CRF data, were removed.

10.1.3. Reports

- A new section was added to describe the contents of the following reports: Performance metrics, SMC, DSMB, and CSR.

10.2 Version 4.0 (current version) contains corrections and additional details requested for Clinical Study Report programming. The blinded statistician continued to not have access to individual level data during these revisions.

10.2.1. Adverse Events of Special Interest

- The appendix describing AESI was inadvertently truncated in Version 3.0. This has been corrected.

10.2.2. Censoring details for Time to Treatment Discontinuation due to Safety

- Definitions for events, competing events, and censored observations were added as Appendix

10.2.3. Description of the per-protocol efficacy analysis

- Censoring definition for per-protocol partial exclusions was added to Section 7.3.1

10.2.4. Revision to the date imputation algorithm for Concomitant and Prior Medications

- A modification was made to section 8.1 to account for the fact that prior medications would have a start date prior to the randomization date.

10.2.5. Description of the per-protocol efficacy analysis

- Censoring definition for per-protocol partial exclusions was added to Section 7.3.1

10.2.6. Clarification of the Injection (Step 2) Efficacy Population

- A modification was made to section 6.5 to state that a participant had to have at least one follow-up visit with HIV status evaluated to be included in this population.

10.2.7. Clarification of Confirmed Pregnancy Population

- Clarification added to Section 6.12 to define additional ways that a pregnancy can be confirmed.

10.2.8. Subgroup definitions

- Section 6.17 was revised. Formerly, the age subgroups were defined as:
"Age: <= 24 vs > 24 years old." This was rewritten as:

“Age:<25 vs >=25years old.”)This would not affect the classification of subjects into age categories.)

- The contraception subgroups were clarified to indicate reference to method during follow-up. Details of efficacy analyses involving this time-dependent covariate were added to Section 7.3.6.
- A note was added to indicate that HSV-2 status at baseline will not be available for the Clinical Study Report. This note is repeated in Section 9.2.

APPENDIX A - Study Visit Windows

HPTN 084 Visit Codes, Target Days, and Visit Windows

HPTN 084 Visit Codes, Target Days, and Visit Windows

Week	Visit Code	Day allowable window opens	Day target window opens	Target Day	Day target window closes	Day allowable window closes
Screening	1.0	--	--	--	--	--
Step 1						
Day 0/Enrollment	2.0	0	0	0	3	6
Week 2	3.0	7	11	14	17	20
Week 4*	4.0	21	25	28	31	31
Step 2*						
Day 0/Week 5*	5.0	0	0	0	3	3
Week 6	6.0	4	4	7	10	17
Week 9	7.0	18	25	28	31	41
Week 13	8.0	42	53	56	59	69
Week 17	9.0	70	81	84	87	97
Week 21	10.0	98	109	112	115	125
Week 25	11.0	126	137	140	143	168
Week 33	12.0	169	189	196	203	224
Week 41	13.0	225	245	252	255	255
Week 42	14.0	256	256	259	266	283
Week 49	15.0	284	301	308	315	336
Week 57	16.0	337	357	364	371	392
Week 65	17.0	393	413	420	427	448
Week 73	18.0	449	469	476	483	504
Week 81	19.0	505	525	532	539	560
Week 89	20.0	561	581	588	595	616
Week 97	21.0	617	637	644	651	672
Week 105	22.0	673	693	700	707	728
Week 113	23.0	729	749	756	763	784
Week 121	24.0	785	805	812	819	840
Week 129	25.0	841	861	868	875	896
Week 137	26.0	897	917	924	931	952
Week 145	27.0	953	973	980	987	1008
Week 153	28.0	1009	1029	1036	1043	1064
Week 161	29.0	1065	1085	1092	1099	1120
Week 169	30.0	1121	1141	1148	1155	1176
Week 177	31.0	1177	1197	1204	1211	1232
Week 185	32.0	1233	1253	1260	1267	1288

Week	Visit Code	Day allowable window opens	Day target window opens	Target Day	Day target window closes	Day allowable window closes
Step 3**						
Day 0 (Step 3 only)	33.0	0	0	<8 weeks from last injection	14	42
Week 12	34.0	43	70	84	98	126
Week 24	35.0	127	154	168	182	210
Week 36	36.0	211	238	252	266	294
Week 48	37.0	295	322	336	350	378

*Please note that the Week 4 and Week 5 Visits must be completed in order for a participant to move to Step 2. If a Week 4 or Week 5 Visit is delayed or missed, contact the CMC for further guidance.

**The target dates for all Step 2 visits are based off of the actual date of the Week 5 Visit. The target dates for all Step 3 visits are based off of the first Step 3 Visit, called "Step 3/Day 0".

Open Label Truvada Schedule						
Week	Visit Code	Day allowable window opens	Day target window opens	Target Day	Day target window closes	Day allowable window closes
Day 0 (date injections permanently discontinue)	V201/or other*	0	0	0	14	42
Week 12	202.0	43	70	84	98	126
Week 24	203.0	127	154	168	182	210
Week 36	204.0	211	238	252	266	294
Week 48	205.0	295	322	336	350	378

- Day 0 for Open Label Truvada Schedule may be a Step 2 visit code or 201. See Section 13.5 Alternate Visits.

Pregnancy Schedule*						
Week	Visit Code	Day allowable window opens	Day target window opens	Target Day	Day target window closes	Day allowable window closes
Day 0 (First positive Pregnancy Test)	XX.X	0	0	0	0	0
4 weeks after first positive pregnancy test	Interim visit XX.X	21	--	28	--	35

Quarterly Visit 1 (12 weeks since first positive pregnancy test)	101	43	70	84	98	126
Quarterly Visit 2 (24 weeks since first positive pregnancy test)	102	127	154	168	182	210
Quarterly Visit 3 (36 weeks since first positive pregnancy test)	103	211	238	252	266	294

*Pregnancy schedule is to be followed throughout pregnancy and while participant is breastfeeding.

Yearly/Annual Visits						
Week	Visit Code	Day allowable window opens	Day target window opens	Target Day	Day target window closes	Day allowable window closes
Day 0 (Last visit participant at clinic HIV Test conducted)	XX.X	0	0	0	0	0
Yearly Visit 1	50.0	182	358	365	372	548
Yearly Visit 2	51.0	547	723	730	737	913
Yearly Visit 3	52.0	912	1088	1095	1102	1278
Yearly Visit 4	53.0	1277	1453	1460	1467	1643
Yearly Visit 5	54.0	1642	1818	1825	1832	2008

Seroconverter Schedule						
Week	Visit Code*	Day allowable window opens	Day target window opens	Target Day	Day target window closes	Day allowable window closes
HIV Confirmatory Visit	XX.X	0	0	0	14	42
Week 12	XX.X	43	70	84	98	126
Week 24	XX.X	127	154	168	182	210
Week 36	XX.X	211	238	252	266	294
Week 48	XX.X	295	322	336	350	378

*Due to unblinding considerations, there are no unique visit codes for seroconverters. The visit codes should reflect the next study visits for the participants.

APPENDIX B - Details of Efficacy Analysis Definitions

Population Definition:

Modified ITT

- Participant is randomized
- Participant is HIV uninfected at randomization (determined by EAC)
- Participant does not have a major enrollment violation – determined by independent adjudication

Note: Only participants with at least one follow-up visit (of any type, including safety and interim visits) with HIV status determined after enrollment will contribute study time to the efficacy analysis.

Censoring for primary efficacy analysis:

- If a study-wide transition to Step 3 occurred, all data from the enrollment visit through the date of transition (plus one month?) will be included
- If no study-wide transition to Step 3 occurred or if the study is stopped early, all ongoing follow-up at the time of decision to stop the blinded portion of the study will be censored on the date of study closure.

Censoring for on blinded study product (OBSP) analysis

Participants will be censored when an injection is delayed, defined as follows:

- **Last non-delayed injection:** The earliest of an injection whose subsequent injection is delayed for the first time (i.e. given >6 weeks after the Week 5 injection or >10 weeks after any other injection) or the last injection before a termination or a permanent product discontinuation
- Follow-up time will be censored at the last visit with HIV status determined up through 6 weeks after the Week 5 injection, if that is the last non-delayed injection, or 10 weeks after the last non-delayed injection for subsequent injections.

Calculation of study time for primary efficacy analysis:

- If HIV infected, study time will be calculated as the number of days between enrollment and the mid-point between first visit where HIV infection was detected and the most recent prior visit where HIV infection was not detectable, as determined by the EAC.
- If HIV uninfected, study time will be calculated as the number of days between enrollment and the last visit with HIV status at or before the censoring time.

Calculation of study time for OBSP analysis:

- If HIV infected prior to OBSP censoring, study time will be calculated as the number of days between enrollment and the mid-point between first visit where HIV infection was detected and the most recent prior visit where HIV infection was not detectable, as determined by the EAC.
- If HIV uninfected At OBSP censoring time, study time will be calculated as the number of days between enrollment and the last visit with HIV status at or before the OBSP censoring time.

Primary endpoint:

- Only infections that are confirmed by the EAC and detected within the analysis timeframe at the time of the final primary analysis adjudication (date of first positive visit as determined by the EAC within the analysis timeframe) will be considered. Infections that are not confirmed by the EAC or that occur out of the analysis timeframe will not be included as primary endpoints.

Interim Analysis (mITT and OBSP) specification

Interim analysis procedures will follow the primary analysis definitions above with the exception that the analysis timeframe will be truncated at the interim-analysis data cut date, as follows.

mITT analysis

- All data from enrollment through the interim analysis cut date will be considered

OBSP analysis

- If there is an injection delay prior to the data cut, all data up to the last visit with HIV status determined that occurred prior to the data cut date and within 6/10 weeks for the first/subsequent injection of the last non-delayed injection will be considered.
- If there is no injection delay prior to the data cut point, data included as for the ITT analysis

APPENDIX C - Definition of Censoring used in analysis of discontinuations due to safety

- Discontinuations due to death, investigator decision and HIV-infection are censored on the discontinuation date
- Participants who exit the study at a scheduled exit visit or at the end of the study are also censored, on the termination date.
- Participants who do not reach the scheduled exit visit and do not discontinue study medications are censored at their last attended visit.
- Permanent discontinuation and/or termination due to participant refusal are treated as competing events, on the discontinuation or termination date, and include: participant unwilling or unable to comply study procedures, injection intolerance, other participant request, or refuse further participation.

The Safety events of interest are:

- Permanent discontinuation due to safety concerns including: clinical/laboratory AE, ISR, CMC recommendation and other clinical reason.

The earliest occurrence of any of the above is selected to determine events versus censored observations versus competing events.

Detailed event/censoring/competing event classifications for reasons for treatment discontinuation or termination:

Events of interest, based on reason for treatment discontinuation:

- CLINICAL AE (PROTOCOL MANDATED)
- LABORATORY AE (PROTOCOL MANDATED)
- CMC RECOMMENDATION BASED ON A CLINICAL EVENT
- CMC RECOMMENDATION BASED ON A LABORATORY VALUE
- CMC RECOMMENDATION FOLLOWING DRUG DISPENSING ERROR
- RASH WHICH THE PARTICIPANT STARTED AFTER THE INITIATION OF INJECTION
- OTHER CLINICAL REASON

Competing events, based on reason for termination:

- PARTICIPANT REFUSED FURTHER PARTICIPATION
- PARTICIPANT RELOCATED, NO FOLLOW-UP PLANNED
- OTHER

Competing events, based on reason for treatment discontinuation:

- PARTICIPANT REQUEST - PARTICIPANT IS UNWILLING OR UNABLE TO COMPLY WITH REQUIRED STUDY PROCEDURES
- PARTICIPANT REQUEST FOR INJECTION INTOLERANCE (AE OR ISR NOT PROTOCOL MANDATED)
- PARTICIPANT REQUEST – OTHER REASON
- PARTICIPANT IS CURRENTLY USING OR PLANNING TO USE PREP OR PEP (OTHER THAN STUDY PRODUCT)
- LOW ORAL ADHERENCE - STEP 1
- PARTICIPANT REFUSED LONG ACTING CONTRACEPTION
- PERCEIVED SIDE EFFECTS OF STUDY DRUG
- POSITIVE PREGNANCY TEST RESULT

Censored observations, based on reason for termination:

- DEATH
- INVESTIGATOR DECISION, SPECIFY
- SCHEDULED EXIT VISIT/END OF STUDY

- HIV INFECTION - STEP 1

Censored observations, based on reason for treatment discontinuation:

- ONE OR MORE REACTIVE HIV TEST RESULTS OR ACUTE HIV INFECTION SUSPECTED
- AS PER CMC: ENROLLMENT ERROR DUE TO DISCORDANT RESULTS (censored on date of randomization)
- HEPATITIS B INFECTION (censored on date of positive test result)

Censored observation, based on loss to follow-up (censored on date of last visit):

(Reason for termination is NULL and Reason for discontinuation is NULL) or Reason for treatment discontinuation = "PARTICIPANT RECEIVED WRONG STUDY PRODUCT AT VISIT 12.0"

Multiple conditions:

Note that a participant may have more than one condition flag indicated for these analyses. The analysis date, ADT, is chosen as the earliest date when multiple conditions are indicated by flag. The event/competing risk/censor variable, CNSR, is chosen as the CNSR value associated with the earliest date when multiple conditions are indicated by flag. If multiple conditions have the same date, the CNSR value is indicated by the earlier value in the following list.

- 1) Safety event,
- 2) Competing events conditions 1, 2, and 3,
- 3) Censoring conditions 1,2, 3, or 4.
- 4) LFU.

Numbers of subjects having had the event of interest, and at risk to have had the event will be shown at selected weekly timepoints under the horizontal axis. The number at risk will be computed as the number of subjects with event times greater than or equal to the given timepoint. The **cumulative** number of events will be computed as the number of subjects with event times **less** than or equal to the event time.

APPENDIX D – Study Sites

Country	City	Site Name	CRS ID
Botswana	Gaborone	Botswana: Gaborone: Gaborone CRS	12701
Kenya	Kisumu	Kenya: Kisumu: Kisumu CRS	31460
Malawi	Blantyre	Malawi: Blantyre: Blantyre CRS	30301
Malawi	Lilongwe	Malawi: Lilongwe: Malawi CRS	12001
South Africa	Botha's Hill	South Africa: Botha's Hill: Botha's Hill CRS	31445
South Africa	Cape Town	South Africa: Cape Town: Emavundleni CRS	30346
South Africa	Cape Town	South Africa: Cape Town: Stellenbosch University (DTTC-SU) CRS	31790
South Africa	Johannesburg	South Africa: Johannesburg: Ward 21	31966
South Africa	Kwa Zulu Natal	South Africa: Kwa Zulu Natal: Isipingo CRS	31635
South Africa	Kwa Zulu Natal	South Africa: Kwa Zulu Natal: Verulam CRS	31663
South Africa	Soweto	South Africa: Soweto: Soweto HPTN CRS	31610
Swaziland	Siteki	Swaziland: Siteki: Swaziland Prevention Center	31994
Uganda	Entebbe	Uganda: Entebbe: UVRI-IAVI	30924
Uganda	Kampala	Uganda: Kampala: Baylor-Uganda CRS	31798
Uganda	Kampala	Uganda: Kampala: MU-JHU Research Collaboration CRS	30293
Zimbabwe	Chitungwiza	Zimbabwe: Chitungwiza: Seke South CRS	30294
Zimbabwe	Chitungwiza	Zimbabwe: Chitungwiza: St.Mary's CRS	30303
Zimbabwe	Chitungwiza	Zimbabwe: Chitungwiza: Zengeza CRS	30320
Zimbabwe	Harare	Zimbabwe: Harare: Parirenyatwa CRS	30313
Zimbabwe	Harare	Zimbabwe: Harare: Spilhaus CRS	30314

APPENDIX E – Adverse Events of Special Interest

Adverse events of special interest (AESI) have been determined for CAB based on pre-clinical and clinical experience, along with information for the integrase inhibitor class of HIV medications safety profile. These are adverse events for which more detailed summaries of event characteristics are of interest. Details of the analyses of AESI are contained in Section **Error! Reference source not found.** based on the current MedDRA version used for the study, v23.1.

The identification and classification of AESI will have occurred as a clinical review of preferred terms for adverse events, lab data, ECG and vital signs prior to the unblinding of the studies. (see Appendix E Section 1).

Table 1 shows the adverse events of special interest identified and reasons for including.

Table 1 Adverse Events of Special Interest

AESI	Product Implicated	Reason for Including as AESI
Hepatic Safety Profile: Assessment of risk of hepatotoxicity	CAB	Clinical, Class, Regulatory Interest,
Hypersensitivity Reactions (HSR)	CAB	Class, Regulatory Interest,
Rash	CAB	Class, Regulatory Interest
Neuropsychiatric Events - Suicidal Ideation/Behaviour - Depression - Bipolar Disorder - Psychosis - Mood Disorders - Anxiety - Sleep Disorders	CAB	Clinical, Class, Regulatory Interest,
Injection Site Reactions (ISR)	CAB	Clinical, Regulatory Interest
Seizures and Seizure-like events	CAB	Clinical, Regulatory Interest
Weight Gain	CAB	Clinical, Class, Regulatory Interest
Rhabdomyolysis	CAB	Clinical, Class, Regulatory Interest
Pancreatitis	CAB	Clinical, Therapeutic Area, Regulatory Interest

Impact on Creatinine	CAB	Regulatory Interest, Therapeutic Area
Safety in Pregnancy (Does not apply to HPTN083)	CAB	Regulatory Interest, Class
Hyperglycaemia/new onset diabetes mellitus	CAB	Observation from 083, Class

Common ISR adverse events are defined by MedDRA preferred terms including injection site pain, injection site induration, injection site nodules and preferred terms of any other ISR with $\geq 5\%$ participants in either treatment arm. ISRs are summarized and listed regardless of IP relationship.

Although not pre-specified as an AESI, further analysis of pyrexia and related terms (Pyrexia Plus), sciatica and specific musculoskeletal related terms, and pancreatitis and related terms (Pancreatitis Plus) will be conducted, based on regulatory interest in the treatment indication, in order to characterise these phenomena in the PrEP population.

The following preferred terms have been identified for these areas of interest:

Pyrexia Plus includes PT: (Chills – 10008531, Feeling of Body Temperature Change - 10061458, Feeling Hot - 10016334, Body Temperature Increased - 10005911, Pyrexia - 10037660, Influenza Like Illness - 10022004, Fever – 10016558).

Sciatica PT: (Sciatica - 10039674) and specific Musculoskeletal includes PT: (Back pain – 10003988, Myalgia – 10028411, Pain in extremity – 10033425, Musculoskeletal pain - 10028391).

Pancreatitis Plus includes PT: (Pancreatitis - 10033645, Pancreatic enzymes increased - 10061900, Abdominal pain - 10000081, Abdominal pain upper - 10000087, Abdominal discomfort - 10000059, Nausea - 10028813). When looking at Pancreatitis Plus events after elevated lipase, the elevated lipase is defined as lipase above the upper limit of normal (ULN).

Section 1: AE Preferred Term for AESI identification (based on MedDRA v23.1)

Hepatotoxicity (Hepatic failure, fibrosis and cirrhosis and other liver damage-related conditions)

PT	PT Code
Acquired hepatocerebral degeneration	10080860
Acute hepatic failure	10000804
Acute on chronic liver failure	10077305
Acute yellow liver atrophy	10070815
Ascites	10003445
Asterixis	10003547
Bacterascites	10068547
Biliary cirrhosis	10004659
Biliary fibrosis	10004664
Cardiohepatic syndrome	10082480

Cholestatic liver injury	10067969
Chronic hepatic failure	10057573
Coma hepatic	10010075
Cryptogenic cirrhosis	10063075
Diabetic hepatopathy	10071265
Drug-induced liver injury	10072268
Duodenal varices	10051010
Flood syndrome	10084797
Gallbladder varices	10072319
Gastric variceal injection	10076237
Gastric variceal ligation	10076238
Gastric varices	10051012
Gastric varices haemorrhage	10057572
Gastroesophageal variceal haemorrhage prophylaxis	10066597
Hepatectomy	10061997
Hepatic atrophy	10019637
Hepatic calcification	10065274
Hepatic cirrhosis	10019641
Hepatic encephalopathy	10019660
Hepatic encephalopathy prophylaxis	10066599
Hepatic failure	10019663
Hepatic fibrosis	10019668
Hepatic hydrothorax	10067365
Hepatic infiltration eosinophilic	10064668
Hepatic lesion	10061998
Hepatic necrosis	10019692
Hepatic steato-fibrosis	10077215
Hepatic steatosis	10019708
Hepatitis fulminant	10019772
Hepatobiliary disease	10062000

Hepatocellular foamy cell syndrome	10053244
Hepatocellular injury	10019837
Hepatopulmonary syndrome	10052274
Hepatorenal failure	10019845
Hepatorenal syndrome	10019846
Hepatotoxicity	10019851
Immune-mediated cholangitis	10083406
Immune-mediated hepatic disorder	10083521
Intestinal varices	10071502
Intestinal varices haemorrhage	10078058
Liver dialysis	10076640
Liver disorder	10024670
Liver injury	10067125
Liver operation	10062040
Liver transplant	10024714
Lupoid hepatic cirrhosis	10025129
Minimal hepatic encephalopathy	10076204
Mixed liver injury	10066758
Nodular regenerative hyperplasia	10051081
Nonalcoholic fatty liver disease	10082249
Non-alcoholic steatohepatitis	10053219
Non-cirrhotic portal hypertension	10077259
Oedema due to hepatic disease	10049631
Oesophageal varices haemorrhage	10030210
Peripancreatic varices	10073215
Portal fibrosis	10074726
Portal hypertension	10036200
Portal hypertensive colopathy	10079446
Portal hypertensive enteropathy	10068923
Portal hypertensive gastropathy	10050897

Portal vein cavernous transformation	10073979
Portal vein dilatation	10073209
Portopulmonary hypertension	10067281
Primary biliary cholangitis	10080429
Regenerative siderotic hepatic nodule	10080679
Renal and liver transplant	10052279
Retrograde portal vein flow	10067338
Reye's syndrome	10039012
Reynold's syndrome	10070953
Splenic varices	10067823
Splenic varices haemorrhage	10068662
Steatohepatitis	10076331
Subacute hepatic failure	10056956
Sugiura procedure	10083010
Varices oesophageal	10056091
Varicose veins of abdominal wall	10072284
White nipple sign	10078438

Hepatotoxicity Continued (Hepatitis, non-infectious)

PT	PT Code
Acute graft versus host disease in liver	10066263
Allergic hepatitis	10071198
Alloimmune hepatitis	10080576
Autoimmune hepatitis	10003827
Chronic graft versus host disease in liver	10072160
Chronic hepatitis	10008909
Graft versus host disease in liver	10064676
Hepatitis	10019717
Hepatitis acute	10019727

Hepatitis cholestatic	10019754
Hepatitis chronic active	10019755
Hepatitis chronic persistent	10019759
Hepatitis fulminant	10019772
Hepatitis toxic	10019795
Immune-mediated hepatitis	10078962
Ischaemic hepatitis	10023025
Lupus hepatitis	10067737
Non-alcoholic steatohepatitis	10053219
Radiation hepatitis	10051015
Steatohepatitis	10076331

Hyperglycaemia/new onset diabetes mellitus

PT	PT Code
Acquired lipoatrophic diabetes	10073667
Blood 1,5-anhydroglucitol decreased	10065367
Blood glucose increased	10005557
Diabetes complicating pregnancy	10012596
Diabetes mellitus	10012601
Diabetes mellitus inadequate control	10012607
Diabetes with hyperosmolarity	10012631
Diabetic arteritis	10077357
Diabetic coma	10012650
Diabetic coronary microangiopathy	10080788
Diabetic hepatopathy	10071265
Diabetic hyperglycaemic coma	10012668
Diabetic hyperosmolar coma	10012669
Diabetic ketoacidosis	10012671
Diabetic ketoacidotic hyperglycaemic coma	10012672
Diabetic ketosis	10012673

Diabetic metabolic decompensation	10074309
Diabetic wound	10081558
Euglycaemic diabetic ketoacidosis	10080061
Fructosamine increased	10017395
Fulminant type 1 diabetes mellitus	10072628
Gestational diabetes	10018209
Glucose tolerance impaired	10018429
Glucose tolerance impaired in pregnancy	10018430
Glucose urine present	10018478
Glycated albumin increased	10082836
Glycosuria	10018473
Glycosuria during pregnancy	10018475
Glycosylated haemoglobin abnormal	10018481
Glycosylated haemoglobin increased	10018484
Hyperglycaemia	10020635
Hyperglycaemic hyperosmolar nonketotic syndrome	10063554
Hyperglycaemic seizure	10071394
Hyperglycaemic unconsciousness	10071286
Impaired fasting glucose	10056997
Insulin resistance	10022489
Insulin resistant diabetes	10022491
Insulin-requiring type 2 diabetes mellitus	10053247
Ketoacidosis	10023379
Ketonuria	10023388
Ketosis	10023391
Ketosis-prone diabetes mellitus	10023392
Latent autoimmune diabetes in adults	10066389
Monogenic diabetes	10075980
Neonatal diabetes mellitus	10028933
New onset diabetes after transplantation	10082630

Pancreatogenous diabetes	10033660
Steroid diabetes	10081755
Type 1 diabetes mellitus	10067584
Type 2 diabetes mellitus	10067585
Type 3 diabetes mellitus	10072659
Urine ketone body present	10057597

Hypersensitivity Reactions Drug reaction with eosinophilia and systemic symptoms syndrome

PT	PT Code
Drug reaction with eosinophilia and systemic symptoms	10073508
Pseudolymphoma	10037127
Granulomatous T-cell pseudolymphoma	10084214
Drug hypersensitivity	10013700
Hypersensitivity	10020751
Type IV Hypersensitivity reaction	10053613
Eosinophillia	10014950
Eye swelling	10015967
Eyelid oedema	10015993
Lip swelling	10024570
Angioedema	10002424
Circumoral oedema	10052250
Face oedema	10016029
Idiopathic angioedema	10073257
Lip oedema	10024558
Mouth swelling	10075203
Oedema mouth	10030110
Periorbital oedema	10034545
Swelling face	10042682
Periorbital swelling	10056647
Swelling of eyelid	10042690

Rash (Severe cutaneous adverse reactions)

PT	PT Code
Acute generalised exanthematous pustulosis	10048799
Bullous haemorrhagic dermatosis	10083809
Cutaneous vasculitis	10011686
Dermatitis bullous	10012441
Dermatitis exfoliative	10012455
Dermatitis exfoliative generalised	10012456
Drug reaction with eosinophilia and systemic symptoms	10073508
Epidermal necrosis	10059284
Erythema multiforme	10015218
Erythrodermic atopic dermatitis	10082985
Exfoliative rash	10064579
Oculomucocutaneous syndrome	10030081
SJS-TEN overlap	10083164
Skin necrosis	10040893
Stevens-Johnson syndrome	10042033
Target skin lesion	10081998
Toxic epidermal necrolysis	10044223
Toxic skin eruption	10057970
Eyelid rash	10074620
Genital rash	10018175
Mucocutaneous rash	10056671
Nodular rash	10075807
Perineal rash	10075364
Rash	10037844
Rash erythematous	10037855
Rash generalised	10037858
Rash macular	10037867

Rash maculo-papular	10037868
Rash maculovesicular	10050004
Rash morbilliform	10037870
Rash papular	10037876
Rash rubelliform	10057984
Rash scarlatiniiform	10037890
Rash vesicular	10037898
Rash pruritic	10037884
Rash follicular	10037857
Rash pustular	10037888
Drug eruption	10013687

Neuropsychiatric Events - consisting of Suicide/self-injury, Depression, Bipolar Disorder, Psychosis and Psychotic Disorders, Mood Disorders, Anxiety, and Sleep Disorders terms will be summarized separately.

Suicide/self-injury

PT	PT Code
Assisted suicide	10079105
Columbia suicide severity rating scale abnormal	10075616
Completed suicide	10010144
Depression suicidal	10012397
Intentional overdose	10022523
Intentional self-injury	10022524
Poisoning deliberate	10036000
Self-injurious ideation	10051154
Suicidal behaviour	10065604
Suicidal ideation	10042458
Suicide attempt	10042464
Suicide threat	10077417
Suspected suicide	10082458
Suspected suicide attempt	10081704

Depression (excl suicide and self injury)

PT	PT Code
Activation syndrome	10066817
Adjustment disorder with depressed mood	10001297
Adjustment disorder with mixed anxiety and depressed mood	10001299
Agitated depression	10001496
Anhedonia	10002511
Antidepressant therapy	10054976
Childhood depression	10068631
Decreased interest	10011971
Depressed mood	10012374
Depression	10012378
Depression postoperative	10012390
Depression rating scale score increased	10084390
Depressive symptom	10054089
Discouragement	10084257
Dysphoria	10013954
Electroconvulsive therapy	10014404
Feeling guilty	10049708
Feeling of despair	10016344
Feelings of worthlessness	10016374
Helplessness	10077169
Major depression	10057840
Menopausal depression	10067371
Mixed anxiety and depressive disorder	10080836
Perinatal depression	10078366
Persistent depressive disorder	10077804
Post stroke depression	10070606
Postictal depression	10071324

Bipolar Disorder

PT	PT Code
Bipolar I disorder	10004939
Bipolar II disorder	10004940
Bipolar disorder	10057667
Cyclothymic disorder	10011724
Hypomania	10021030
Mania	10026749
Manic symptom	10084119

Psychosis and psychotic disorders

PT	PT Code
Acute psychosis	10001022
Alcoholic psychosis	10001632
Alice in wonderland syndrome	10001666
Brief psychotic disorder with marked stressors	10048549
Brief psychotic disorder without marked stressors	10056395
Brief psychotic disorder, with postpartum onset	10006362
Charles Bonnet syndrome	10063354
Childhood psychosis	10061040
Clang associations	10009232
Cotard's syndrome	10059591
Delusion	10012239
Delusion of grandeur	10012241
Delusion of parasitosis	10012242
Delusion of reference	10012244
Delusion of replacement	10012245
Delusion of theft	10084030

Delusional disorder, erotomanic type	10012249
Delusional disorder, grandiose type	10012250
Delusional disorder, jealous type	10012251
Delusional disorder, mixed type	10012252
Delusional disorder, persecutory type	10053195
Delusional disorder, somatic type	10012254
Delusional disorder, unspecified type	10012255
Delusional perception	10012258
Dementia of the Alzheimer's type, with delusions	10012295
Depressive delusion	10063033
Derailment	10012411
Epileptic psychosis	10059232
Erotomanic delusion	10015134
Flight of ideas	10016777
Hallucination	10019063
Hallucination, auditory	10019070
Hallucination, gustatory	10019071
Hallucination, olfactory	10019072
Hallucination, synaesthetic	10062824
Hallucination, tactile	10019074
Hallucination, visual	10019075
Hallucinations, mixed	10019079
Hypnagogic hallucination	10020927

Hypnopompic hallucination	10020928
Hysterical psychosis	10062645
Ideas of reference	10021212
Illusion	10021403
Jealous delusion	10023164
Loose associations	10024825
Mixed delusion	10076429
Neologism	10028916
Neuroleptic-induced deficit syndrome	10075295
Paranoia	10033864
Paranoid personality disorder	10033869
Parkinson's disease psychosis	10074835
Paroxysmal perceptual alteration	10063117
Persecutory delusion	10034702
Postictal psychosis	10070669
Post-injection delirium sedation syndrome	10072851
Posturing	10036437
Psychosis postoperative	10065617
Psychotic behaviour	10037249
Psychotic disorder	10061920
Psychotic disorder due to a general medical condition	10061921
Reactive psychosis	10053632
Rebound psychosis	10074833

Schizoaffective disorder	10039621
Schizoaffective disorder bipolar type	10068889
Schizoaffective disorder depressive type	10068890
Schizophrenia	10039626
Schizophreniform disorder	10039647
Schizotypal personality disorder	10039651
Senile psychosis	10039987
Shared psychotic disorder	10040535
Somatic delusion	10041317
Somatic hallucination	10062684
Substance-induced psychotic disorder	10072388
Tangentiality	10043114
Thought blocking	10043495
Thought broadcasting	10052214
Thought insertion	10043496
Thought withdrawal	10043497
Transient psychosis	10056326
Waxy flexibility	10047853

Mood Disorders

PT	PT Code
Affect lability	10054196
Affective ambivalence	10077173
Affective disorder	10001443
Alexithymia	10077719

Anger	10002368
Apathy	10002942
Blunted affect	10005885
Boredom	10048909
Constricted affect	10010778
Crying	10011469
Diencephalic syndrome of infancy	10012774
Discouragement	10084257
Dysphoria	10013954
Emotional disorder	10014551
Emotional distress	10049119
Emotional poverty	10014557
Euphoric mood	10015535
Flat affect	10016759
Frustration tolerance decreased	10077753
Inappropriate affect	10021588
Irritability	10022998
Laziness	10051602
Lethargy	10024264
Listless	10024642
Moaning	10027783
Mood altered	10027940
Mood disorder due to a general medical condition	10027944
Mood swings	10027951
Morose	10027977
Neuroleptic-induced deficit syndrome	10075295
Premenstrual dysphoric disorder	10051537
Premenstrual syndrome	10036618
Screaming	10039740
Seasonal affective disorder	10039775
Steroid withdrawal syndrome	10042028
Substance-induced mood disorder	10072387

Anxiety

PT	PT Code
Acrophobia	10000605
Activation syndrome	10066817
Acute stress disorder	10001084
Aerophobia	10080300
Agitation	10001497
Agitation postoperative	10049989
Agoraphobia	10001502
Akathisia	10001540
Algophobia	10078056
Animal phobia	10002518
Anniversary reaction	10074066
Anticipatory anxiety	10002758
Anxiety	10002855
Anxiety disorder	10057666
Anxiety disorder due to a general medical condition	10002859
Arachnophobia	10051408
Astraphobia	10078372
Autophobia	10071070
Body dysmorphic disorder	10052793
Burnout syndrome	10065369
Catastrophic reaction	10082329
Cibophobia	10082413
Claustrophobia	10009244
Compulsions	10010219
Compulsive cheek biting	10076510
Compulsive handwashing	10071263
Compulsive hoarding	10068007
Compulsive lip biting	10066241
Compulsive shopping	10067948
Cryophobia	10082662
Dermatillomania	10065701
Dysmorphophobia	10049096

Emetophobia	10070637
Fear	10016275
Fear of animals	10016276
Fear of closed spaces	10016277
Fear of crowded places	10050365
Fear of death	10066392
Fear of disease	10016278
Fear of eating	10050366
Fear of falling	10048744
Fear of injection	10073753
Fear of open spaces	10016279
Fear of pregnancy	10067035
Fear of surgery	10084519
Fear of weight gain	10016280
Fear-related avoidance of activities	10080136
Generalised anxiety disorder	10018075
Glossophobia	10080077
Haemophobia	10073458
Haphephobia	10067580
Herpetophobia	10081809
Hydrophobia	10053317
Hyperarousal	10080831
Immunisation anxiety related reaction	10075205
Kinesiophobia	10078430
Limited symptom panic attack	10024511
Mysophobia	10078769
Nail picking	10066779
Nervousness	10029216
Neurosis	10029333
Noctiphobia	10057946
Nocturnal fear	10057948
Nosocomophobia	10083993
Nosophobia	10063546
Obsessive need for symmetry	10077179

Obsessive rumination	10056264
Obsessive thoughts	10029897
Obsessive-compulsive disorder	10029898
Obsessive-compulsive symptom	10077894
Ochlophobia	10050095
Osmophobia	10060765
Paediatric autoimmune neuropsychiatric disorders associated with streptococcal infection	10072147
Panic attack	10033664
Panic disorder	10033666
Panic reaction	10033670
Paruresis	10069024
Performance fear	10034432
Phagophobia	10050096
Pharmacophobia	10069423
Phobia	10034912
Phobia of driving	10056676
Phobia of exams	10034913
Phobic avoidance	10034918
Phonophobia	10054956
Photiaugiaphobia	10064420
Postpartum anxiety	10082233
Postpartum neurosis	10036419
Postpartum stress disorder	10056394
Post-traumatic stress disorder	10036316
Procedural anxiety	10075204
Pseudoangina	10056610
Selective mutism	10039917
Separation anxiety disorder	10040045
Sitophobia	10080170
Social anxiety disorder	10041242
Social fear	10041247
Stress	10042209
Tension	10043268

Terminal agitation	10077416
Thanatophobia	10064723
Thermophobia	10075147
Trichotemnomania	10072752
Trichotillomania	10044629

Sleep Disorders

PT	PT Code
Abnormal dreams	10000125
Abnormal sleep-related event	10061613
Advanced sleep phase	10001423
Behavioural induced insufficient sleep syndrome	10081938
Behavioural insomnia of childhood	10072072
Breathing-related sleep disorder	10006344
Cataplexy	10007737
Circadian rhythm sleep disorder	10009191
Confusional arousal	10067494
Delayed sleep phase	10012209
Dyssomnia	10061827
Exploding head syndrome	10080684
Hypersomnia	10020765
Hypersomnia related to another mental condition	10020767
Hypersomnia-bulimia syndrome	10053712
Hypnagogic hallucination	10020927
Hypnopompic hallucination	10020928
Hyposomnia	10067530
Initial insomnia	10022035
Insomnia	10022437
Insomnia related to another mental condition	10022443
Irregular sleep phase	10022995
Irregular sleep wake rhythm disorder	10080301
Loss of dreaming	10065085
Middle insomnia	10027590
Narcolepsy	10028713

Nightmare	10029412
Non-24-hour sleep-wake disorder	10078086
Parasomnia	10061910
Paradoxical insomnia	10083337
Periodic limb movement disorder	10064600
Pickwickian syndrome	10035004
Poor quality sleep	10062519
Rapid eye movement sleep behaviour disorder	10077299
Rapid eye movements sleep abnormal	10037841
Shift work disorder	10078088
Sleep apnoea syndrome	10040979
Sleep attacks	10040981
Sleep disorder	10040984
Sleep disorder due to a general medical condition	10063910
Sleep disorder due to general medical condition, hypersomnia type	10040985
Sleep disorder due to general medical condition, insomnia type	10040986
Sleep disorder due to general medical condition, mixed type	10040987
Sleep disorder due to general medical condition, parasomnia type	10040988
Sleep inertia	10067493
Sleep paralysis	10041002
Sleep sex	10067492
Sleep talking	10041009
Sleep terror	10041010
Sleep-related eating disorder	10067315
Somnambulism	10041347
Somnolence	10041349
Somnolence neonatal	10041350
Sopor	10058709
Stupor	10042264
Terminal insomnia	10068932
Upper airway resistance syndrome	10063968

Injection site Reactions

Use CRF terms for ISR.

Safety in Pregnancy (084 only)

Use AE terms co-reported in pregnancy exposures to CAB.

Seizures/Convulsions

PT	PT Code
1p36 deletion syndrome	10082398
2-Hydroxyglutaric aciduria	10078971
Acquired epileptic aphasia	10052075
Acute encephalitis with refractory, repetitive partial seizures	10076948
Alcoholic seizure	10056347
Alpers disease	10083857
Aspartate-glutamate-transporter deficiency	10079140
Atonic seizures	10003628
Atypical benign partial epilepsy	10056699
Automatism epileptic	10003831
Autonomic seizure	10049612
Baltic myoclonic epilepsy	10054895
Benign familial neonatal convulsions	10067866
Benign rolandic epilepsy	10070530
Biotinidase deficiency	10071434
CEC syndrome	10083749
CDKL5 deficiency disorder	10083005
Change in seizure presentation	10075606
Clonic convulsion	10053398
Congenital bilateral perisylvian syndrome	10082716

Convulsion in childhood	10052391
Convulsions local	10010920
Convulsive threshold lowered	10010927
CSWS syndrome	10078827
Deja vu	10012177
Double cortex syndrome	10073490
Dreamy state	10013634
Drug withdrawal convulsions	10013752
Early infantile epileptic encephalopathy with burst-suppression	10071545
Eclampsia	10014129
Epilepsy	10015037
Epilepsy surgery	10079824
Epilepsy with myoclonic-atonic seizures	10081179
Epileptic aura	10015049
Epileptic psychosis	10059232
Faciobrachial dystonic seizure	10084187
Febrile convulsion	10016284
Febrile infection-related epilepsy syndrome	10079438
Focal dyscognitive seizures	10079424
Frontal lobe epilepsy	10049424
Gelastictic seizure	10082918
Generalised onset non-motor seizure	10083376
Generalised tonic-clonic seizure	10018100

Glucose transporter type 1 deficiency syndrome	10078727
GM2 gangliosidosis	10083933
Grey matter heterotopia	10082084
Hemimegalencephaly	10078100
Hyperglycaemic seizure	10071394
Hypocalcaemic seizure	10072456
Hypoglycaemic seizure	10048803
Hyponatraemic seizure	10073183
Idiopathic generalised epilepsy	10071081
Infantile spasms	10021750
Jeavons syndrome	10084303
Juvenile myoclonic epilepsy	10071082
Lafora's myoclonic epilepsy	10054030
Lennox-Gastaut syndrome	10048816
Migraine-triggered seizure	10076676
Molybdenum cofactor deficiency	10069687
Multiple subpial transection	10079825
Myoclonic epilepsy	10054859
Myoclonic epilepsy and ragged-red fibres	10069825
Neonatal epileptic seizure	10082068
Neonatal seizure	10082067
Partial seizures	10061334
Partial seizures with secondary generalisation	10056209

Petit mal epilepsy	10034759
Polymicrogyria	10073489
Post stroke epilepsy	10076982
Post stroke seizure	10076981
Postictal headache	10052470
Postictal paralysis	10052469
Postictal psychosis	10070669
Postictal state	10048727
Post-traumatic epilepsy	10036312
Schizencephaly	10073487
Seizure	10039906
Seizure anoxic	10039907
Seizure cluster	10071350
Seizure like phenomena	10071048
Severe myoclonic epilepsy of infancy	10073677
Simple partial seizures	10040703
Status epilepticus	10041962
Sudden unexplained death in epilepsy	10063894
Temporal lobe epilepsy	10043209
Tonic clonic movements	10051171
Tonic convulsion	10043994
Tonic posturing	10075125
Topectomy	10073488

Transient epileptic amnesia	10081728
Tuberous sclerosis complex	10080584
Uncinate fits	10045476
Confusional state	10010305
Loss of consciousness	10024855
Syncope	10042772
Sopor	10058709
Stupor	10042264
Altered state of consciousness	10050093
Depressed level of consciousness	10012373
Consciousness fluctuating	10050093

Weight Gain

PT	PT Code
Abdominal fat apron	10077983
Overweight	10033307
Abnormal weight gain	10000188
Central obesity	10065941
Obesity	10029883
Weight abnormal	10056814
Weight increased	10047899
Waist circumference increased	10064863
Body mass index abnormal	10074506
Body mass index increased	10005897
Fat tissue increased	10016251
Sarcopenic obesity	10083992

Rhabdomyolysis

PT	PT Code
Muscle necrosis	10028320
Myoglobin blood increased	10028625
Myoglobin blood present	10059888
Myoglobin urine present	10028631
Myoglobinaemia	10058735
Myoglobinuria	10028629
Myopathy	10028641
Myopathy toxic	10028648
Necrotising myositis	10074769
Rhabdomyolysis	10039020
Thyrotoxic myopathy	10081524
Myalgia	10028411
Myositis	10028653

Pancreatitis

PT	PT Code
Cullen's sign	10059029
Grey Turner's sign	10075426
Haemorrhagic necrotic pancreatitis	10076058
Hereditary pancreatitis	10056976
Immune-mediated pancreatitis	10083072
Ischaemic pancreatitis	10066127
Oedematous pancreatitis	10052400
Pancreatic abscess	10048984
Pancreatic cyst drainage	10082531

Pancreatic haemorrhage	10033625
Pancreatic necrosis	10058096
Pancreatic phlegmon	10056975
Pancreatic pseudoaneurysm	10081762
Pancreatic pseudocyst	10033635
Pancreatic pseudocyst drainage	10033636
Pancreatic pseudocyst haemorrhage	10083813
Pancreatic pseudocyst rupture	10083811
Pancreatitis	10033645
Pancreatitis acute	10033647
Pancreatitis haemorrhagic	10033650
Pancreatitis necrotising	10033654
Pancreatitis relapsing	10033657
Pancreatorenal syndrome	10056277
Subacute pancreatitis	10084554

Impact on Creatinine

PT	PT Code
Acute kidney injury	10069339
Acute phosphate nephropathy	10069688
Anuria	10002847
Azotaemia	10003885
Continuous haemodiafiltration	10066338
Dialysis	10061105
Foetal renal impairment	10078987
Haemodialysis	10018875
Haemofiltration	10053090

Neonatal anuria	10049778
Nephropathy toxic	10029155
Oliguria	10030302
Peritoneal dialysis	10034660
Prerenal failure	10072370
Renal failure	10038435
Renal failure neonatal	10038447
Renal impairment	10062237
Renal impairment neonatal	10049776
Subacute kidney injury	10081980
Atypical haemolytic uraemic syndrome	10079840
Cardiorenal syndrome	10068230
Chronic kidney disease	10064848
Crush syndrome	10050702
Diabetic end stage renal disease	10012660
End stage renal disease	10077512
Foetal renal impairment	10078987
Haemolytic uraemic syndrome	10018932
Hepatorenal failure	10019845
Hepatorenal syndrome	10019846
Nail-patella syndrome	10063431
Neonatal anuria	10049778
Oliguria	10030302
Pancreatorenal syndrome	10056277
Postoperative renal failure	10056675
Postrenal failure	10059345
Prerenal failure	10072370
Propofol infusion syndrome	10063181
Renal failure	10038435
Renal failure neonatal	10038447
Renal impairment	10062237
Renal impairment neonatal	10049776
Renal injury	10061481
Scleroderma renal crisis	10062553

Traumatic anuria	10044501
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Section 2: Events of Interest in Relation to Injections

Pyrexia Plus

PT	PT Code
Body temperature increased	10005911
Chills	10008531
Feeling of body temperature change	10061458
Feeling hot	10016334
Pyrexia	10037660
Influenza like illness	10022004
Fever	10016558

Sciatica

PT	PT Code
Sciatica	10039674

Musculoskeletal Events

PT	PT Code
Back pain	10003988
Myalgia	10028411
Pain in extremity	10033425
Musculoskeletal pain	10028391

Section 3: Additional Analysis of Pancreatitis

Pancreatitis Plus

PT	PT Code
Pancreatitis	10033645
Pancreatic enzymes increased	10061900
Abdominal pain	10000081
Abdominal pain upper	10000087
Abdominal discomfort	10000059

Nausea	10028813
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